

NxM Templates for SuperCDMS SNOLAB Run 4 — Ge Activation Data

K-line event selection · free-pretrigger two-exponential fits (*future step: three-, four-exponential fits*) · 1x1
and NxM (PCA) templates

Zhiheng Li — July 2026

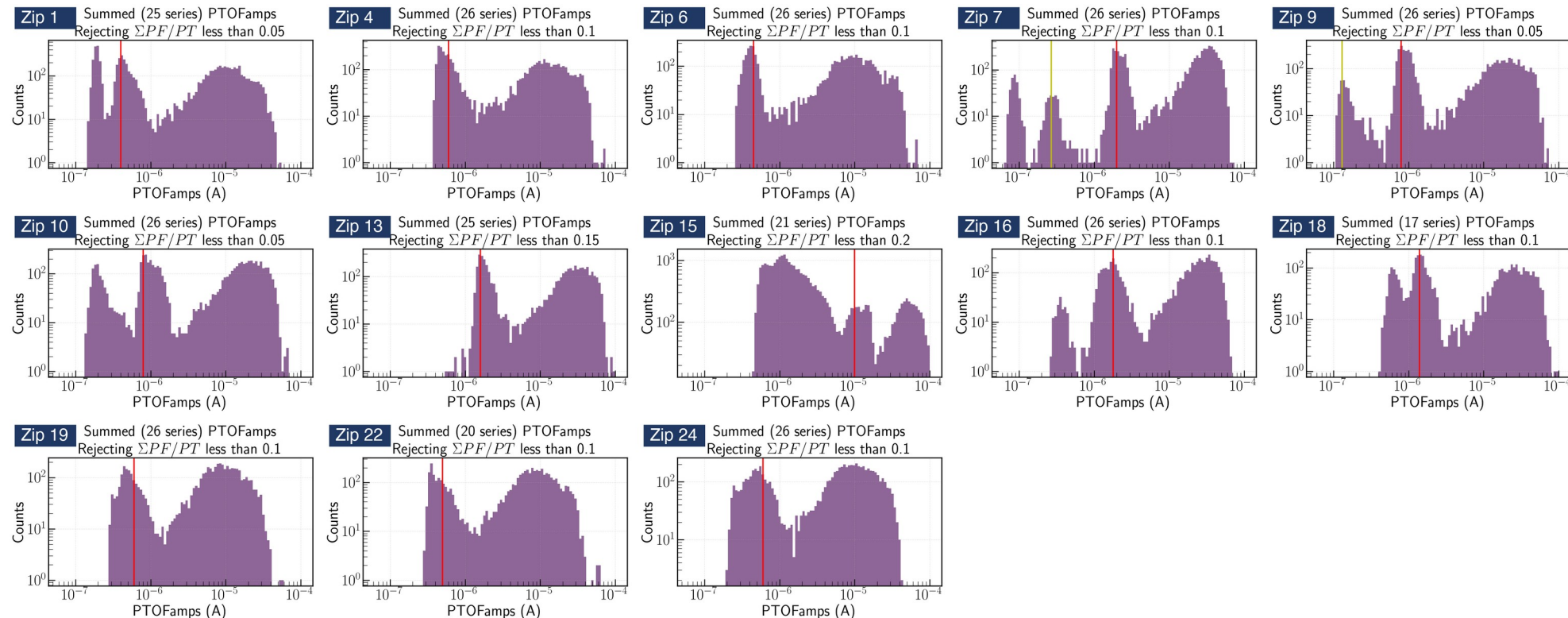
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4. **Cross-checks:** cuts remove noise, real pulses stay; slow-fall follow-up
5. **Template generation:** 1x1 weighted template, NxM PCA templates
6. **Deliverables and future steps:** cdmsbats ROOT files; three-, four-exponential fits
7. **Backup:** weak detectors, rise-time cut details, full results for every zip

Ge activation dataset - R4

Credit: Prof. Tarek Saab

- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in **PTOFamps** (the red line in each panel below)
- A **PTOFamps** window bracketing the red line: position $\times/\div 1.35$, a rough eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata, saved as per-series pickle files (reusable for any later analysis)
- Events in the window per zip: Z1 10.1k Z4 26.1k Z6 17.0k Z7 2.2k Z9 2.6k Z10 7.6k Z13 9.1k Z15 4.2k Z16 1.6k Z18 22.1k Z19 23.3k Z22 25.0k Z24 22.9k (total 174k)
- **This talk focuses on Z7, the best detector; all other detectors are covered in the backup slides**

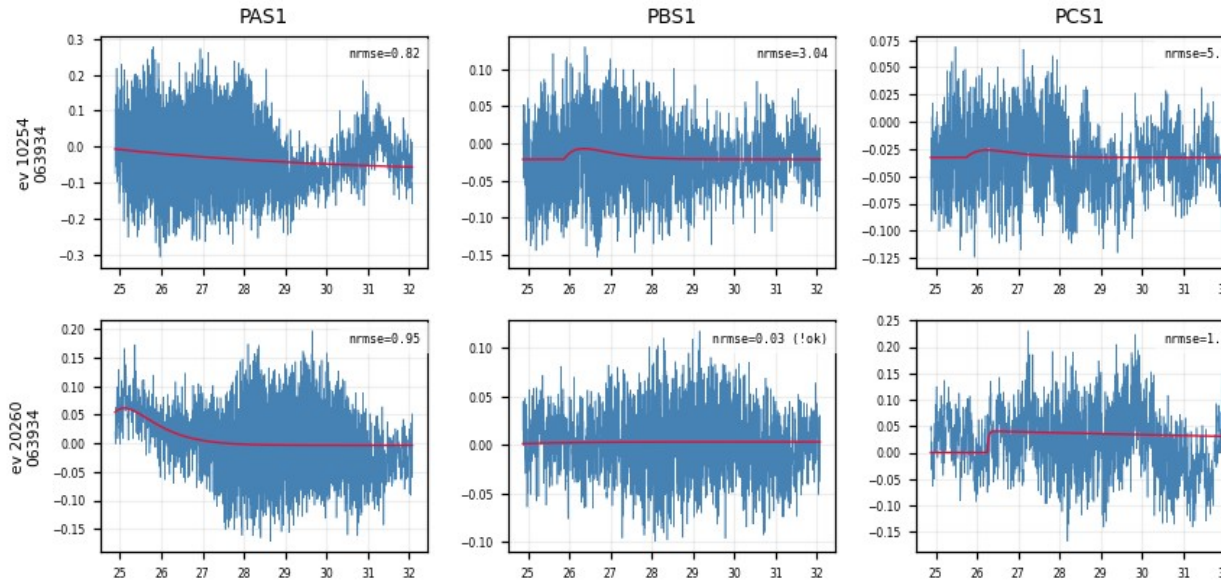


All 13 detectors, PTOFamps spectra (Prof. Saab's study)

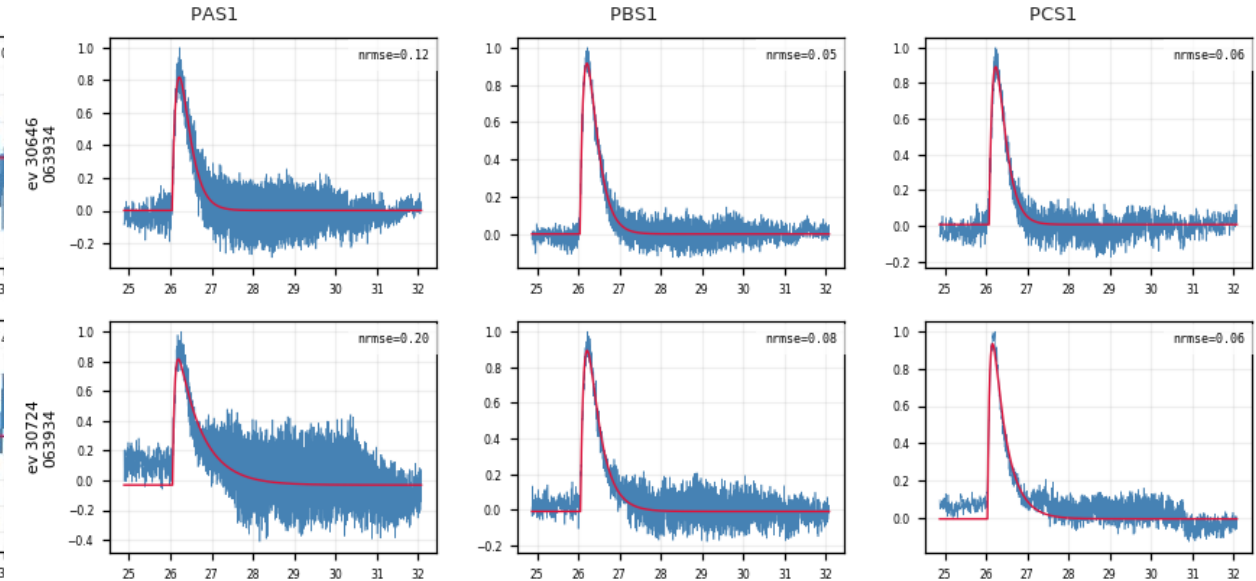
Fit quality at a glance: event × channel grids

- Real events and noise triggers can be told apart by visual comparison (no cut applied at this stage)

NOISE



REAL PULSE



Per-trace algorithm: low-pass → fit → align

1. **Low-pass** — 100 kHz Butterworth
2. **Normalize** — subtract the baseline (median of the pre-pulse samples), scale the trace to unit peak
3. **Fit** — two-exponential model:

$$y(t) = A[e^{-(t - t_0)/\tau_{\text{fall}}} - e^{-(t - t_0)/\tau_{\text{rise}}}] + b$$

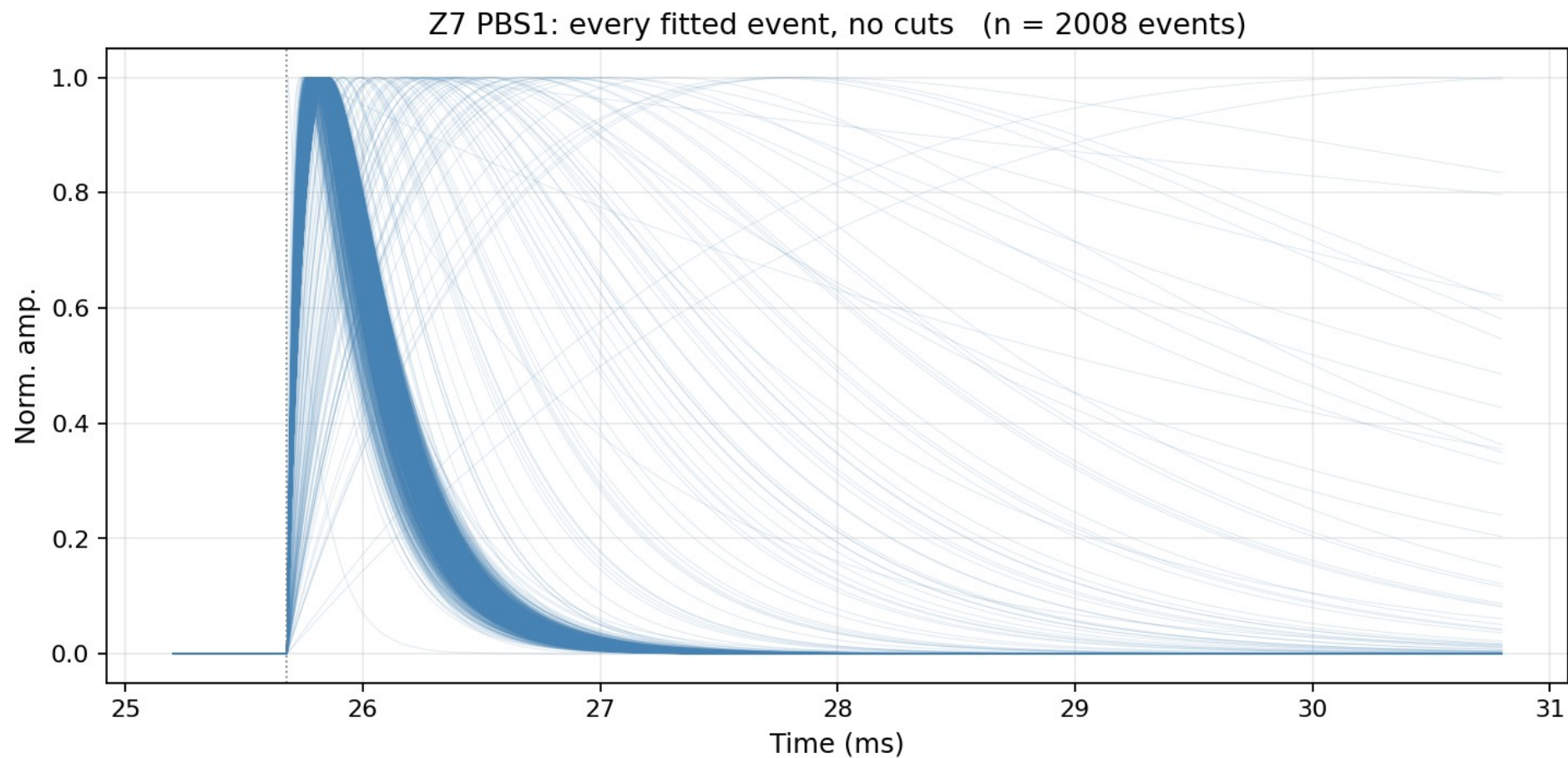
t_0 = pretrigger, free within 16050 ± 3000 samples (τ_{rise} , τ_{fall} : rise / fall time)

Fit parameters: A amplitude · τ_{rise} rise time constant · τ_{fall} fall time constant · t_0 pulse onset (pretrigger) · b baseline

4. **Two quality checks per trace** (calculated and recorded here, not cut yet):

- **fit_ok** — amplitude is positive, and the pulse rises faster than it falls
- **NRMSE** — RMS of the fit residual ÷ pulse peak (smaller = better fit)

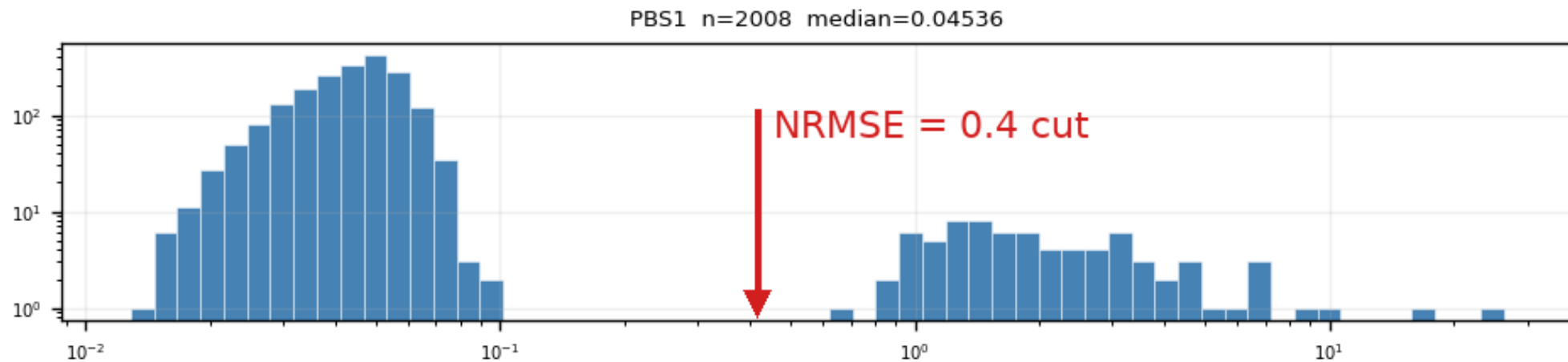
5. Align



Z7 PBS1: every fitted event drawn at the common pretrigger, no cuts of any kind (n = 2008 events)

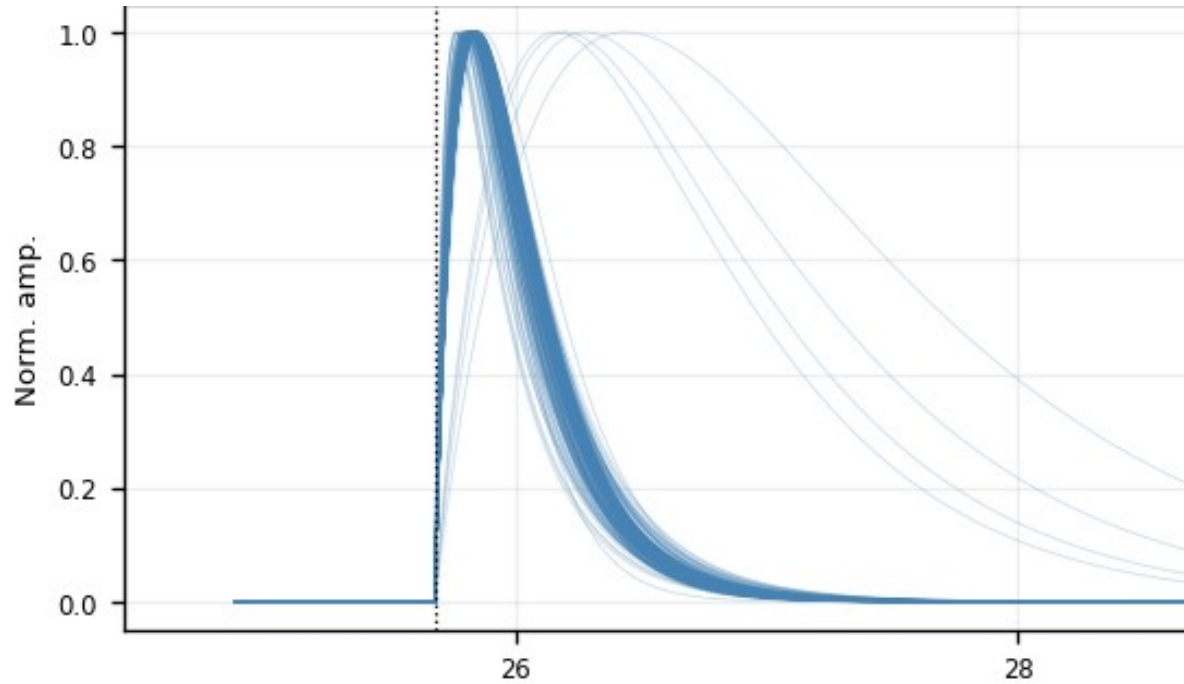
Quality cut 1: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05 – 0.1) vs noise triggers (≈ 1 – 2), valley at ≈ 0.4 – 0.5
- The cut sits in the valley: eyeball read off the distribution
- Worse detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

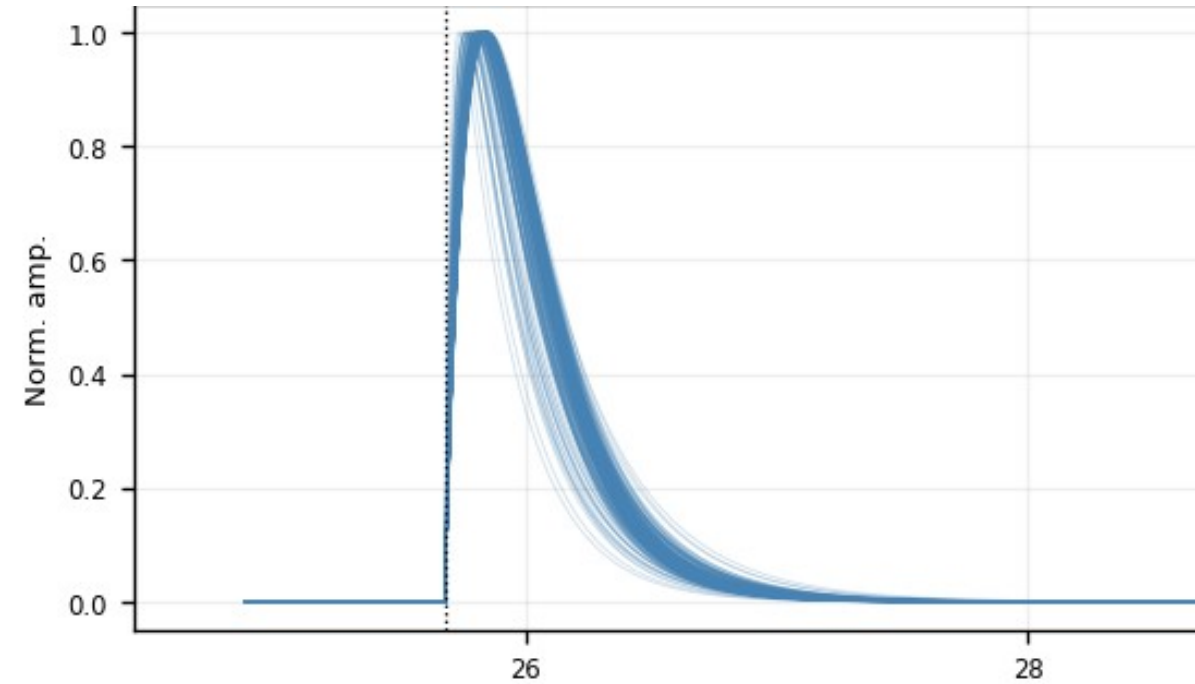


Z7 PBS1: two clean populations, log-log axes

Aligned fitted curves: before and after the NRMSE cut



All fit_ok fitted curves (Z7 PBS1), 200-curve sample drawn

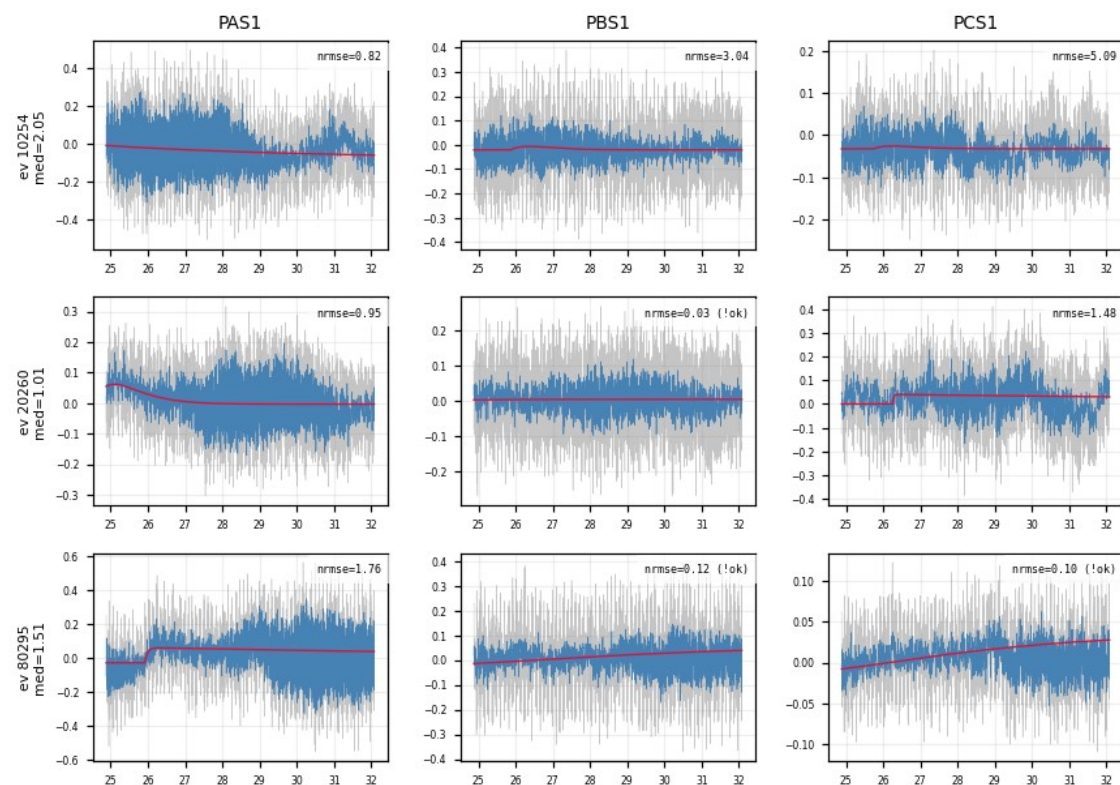


After $\text{NRMSE} \leq 0.4$ (Z7 PBS1): one tight shape family

The rejected population is noise — verified in the raw traces

- Example of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse
- the cut removes noise triggers, not physics

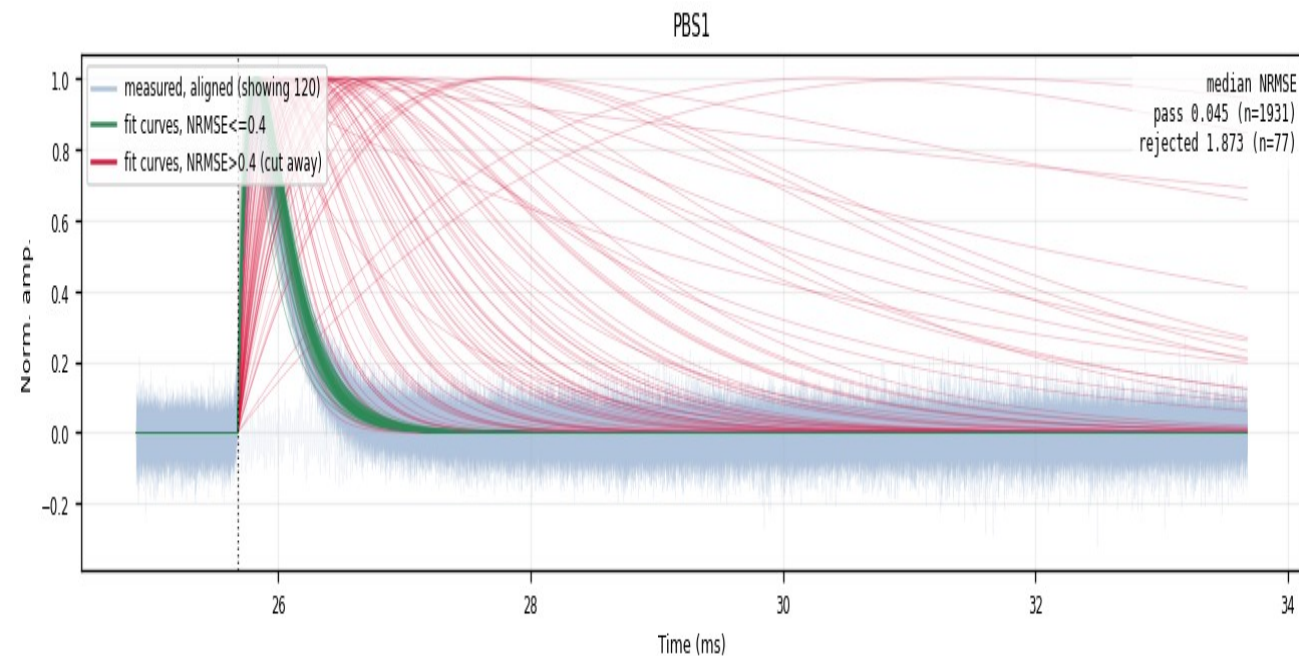
Rejected events: raw vs fit



Z7: NRMSE-rejected events — raw traces are noise

Aligned curves: kept vs rejected

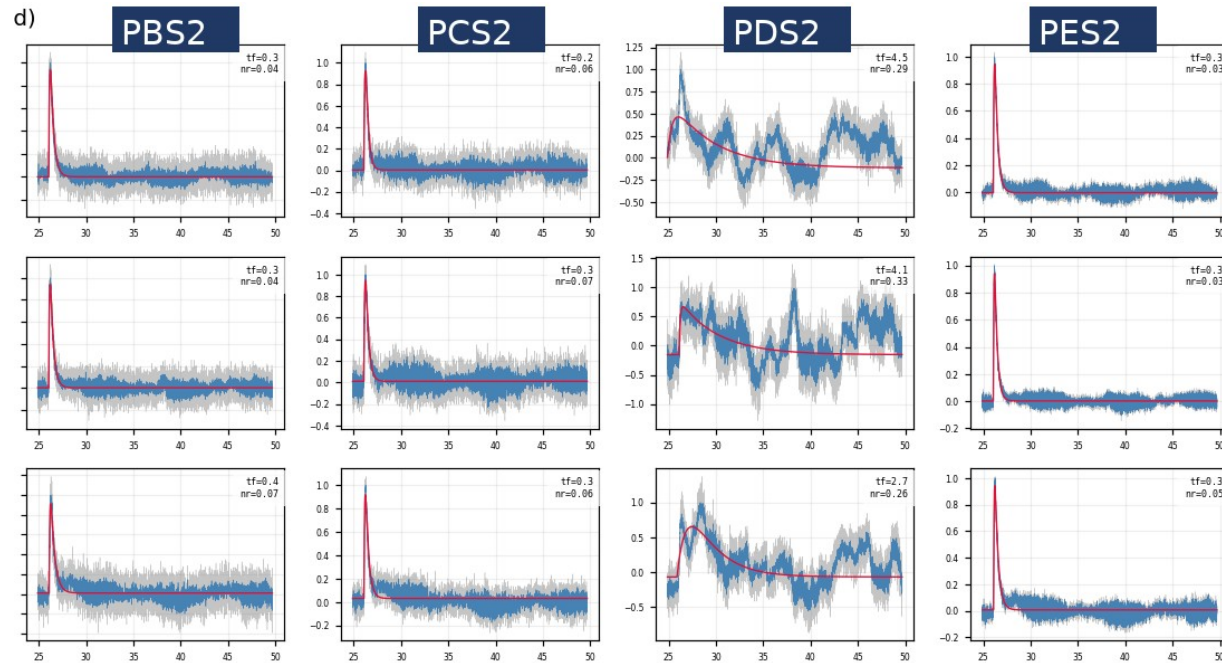
median NRMSE: **kept 0.045 (n = 1931)** · **rejected 1.873 (n = 77)**



Z7 PBS1 — **passes the cut (kept)** — **cut away = rejected**
faint blue = measured traces 1931 kept / 77 cut (~3.8%)

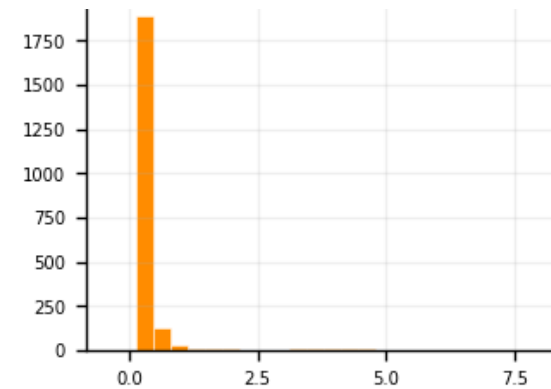
Slow-fall tail in one particular channel

- The post-cut plot still shows slow-fall tails → sample them: $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{fall}} > 1.5$ ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses. Their extreme fall times trace to one channel: only PDS2 swings (τ_{fall} median 0.51 ms vs ≈ 0.25 ms elsewhere)
- Conclusion: no τ_{fall} cut — slow-fall events passing NRMSE stay in
- PDS2 alone: can not make useful templates; hard to fix in analysis/software, the noise itself needs to be fixed
- Temporary solution: use the PDS1 template in place of PDS2 (applied in the delivered ROOT files)

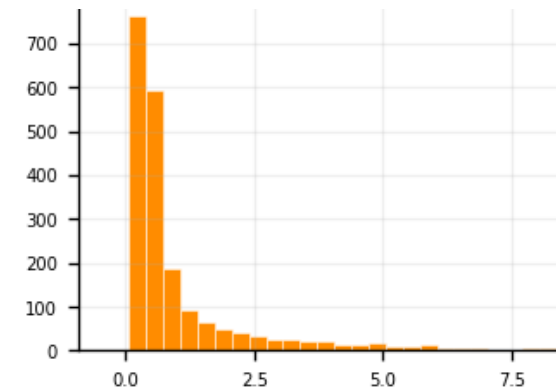


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_{fall} distribution, same 0–7 ms axis:



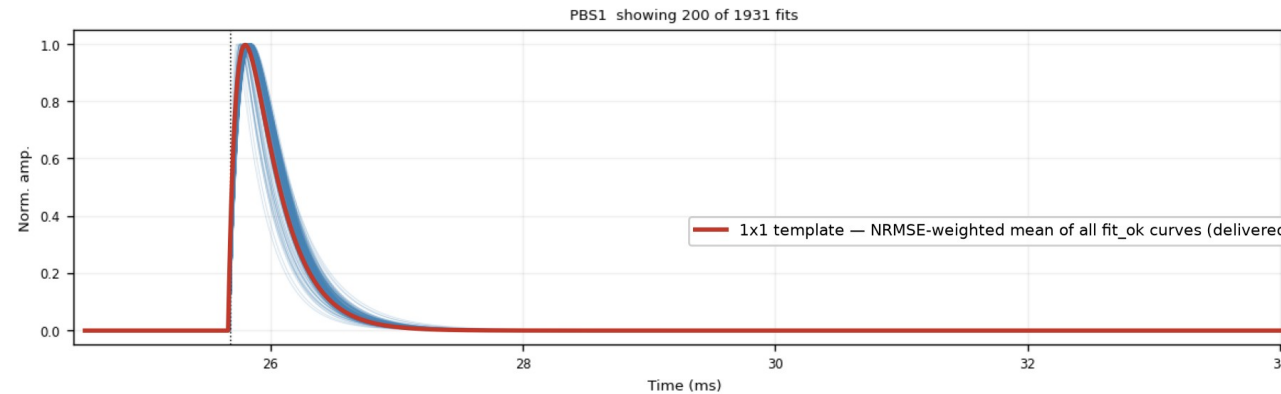
Z7 PAS1 — normal channel: narrow
(median 0.25 ms)



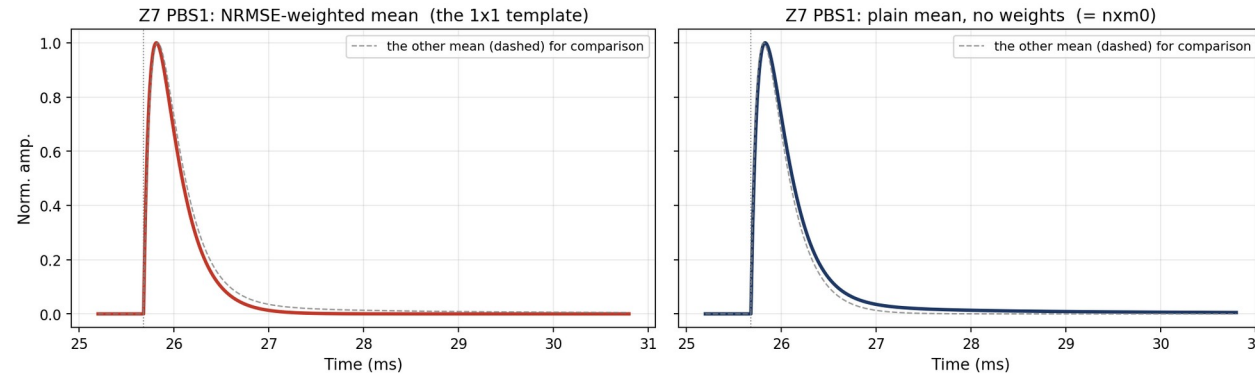
Z7 PDS2 — bad channel: broad tail
(median 0.51 ms)

Results: 1x1 Templates

- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction



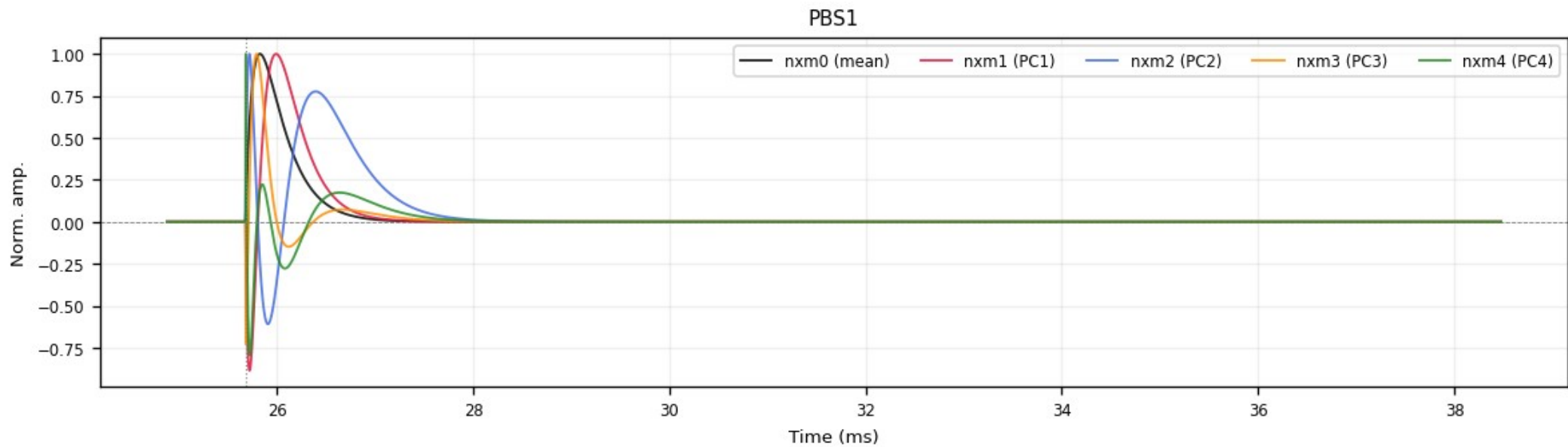
Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the 1x1 template



Same curves, two averages (zoomed): NRMSE-weighted (the 1x1 template) vs plain mean (= nxm0); dashed gray = the other one

Results: NxM Templates

- channel-specific PCA over the fitted curves ($\text{fit_ok} + \text{NRMSE} \leq 0.4$)
- nxm0 = mean shape; nxm1 – 4 = principal components; $\text{real pulse} = \sum_i \text{amp}_i \cdot \text{nxm}_i$
- $\text{PC1} + \text{PC2} \approx 96\text{--}98\%$ of the shape variance



Z7 PBS1: nxm0 (mean) + nxm1 – 4 (PCs) — the oscillating components encode rise/fall-time variation

Template file

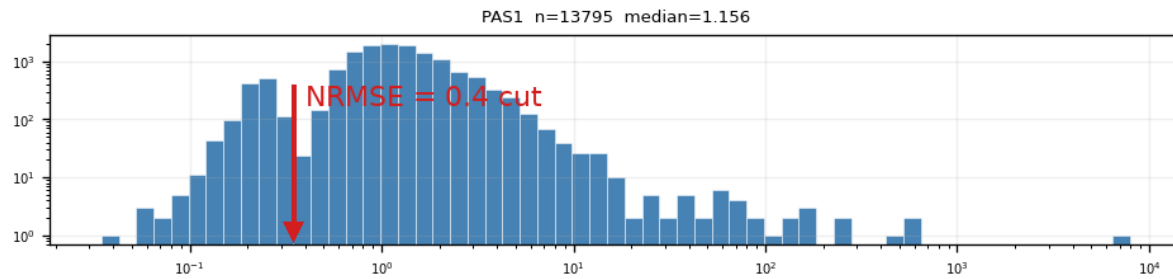
- Two template sets for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- In cdmsbats_config feature branch, ready for merge

Future Steps

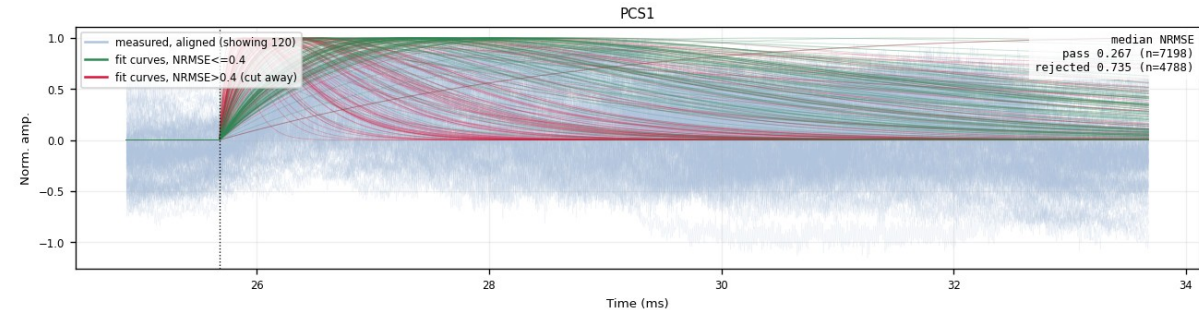
- Three-, four-exponential fits
- Exercise NxM processing
- Other detectors: worse noise, a few more cuts added, in the backup slides

Backup — weak detectors (example: Z22)

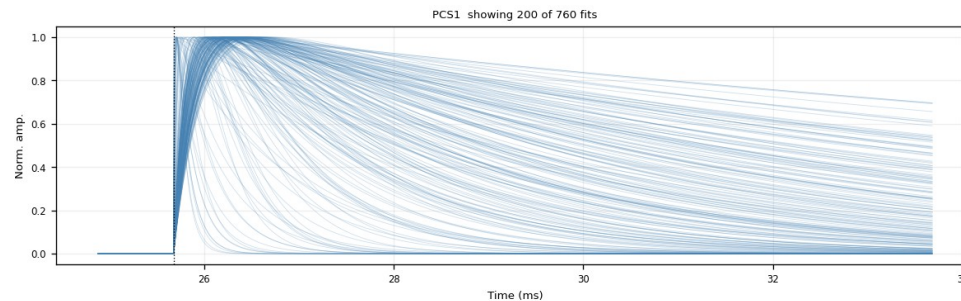
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line sits inside the noise population — the PTOF window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominates; the same 0.4 cut digs the real pulses out — Z22 PCS1: 7198 kept / 4788 cut (~40%)
- Kept bundle broader than on quiet detectors; steep red curves = fits latching onto sharp noise spikes, not fast pulses being cut
- $\tau_{\text{rise}} \leq 0.3$ ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7 — residual slow drift



Z22 PAS1: NRMSE histogram — noise dominates



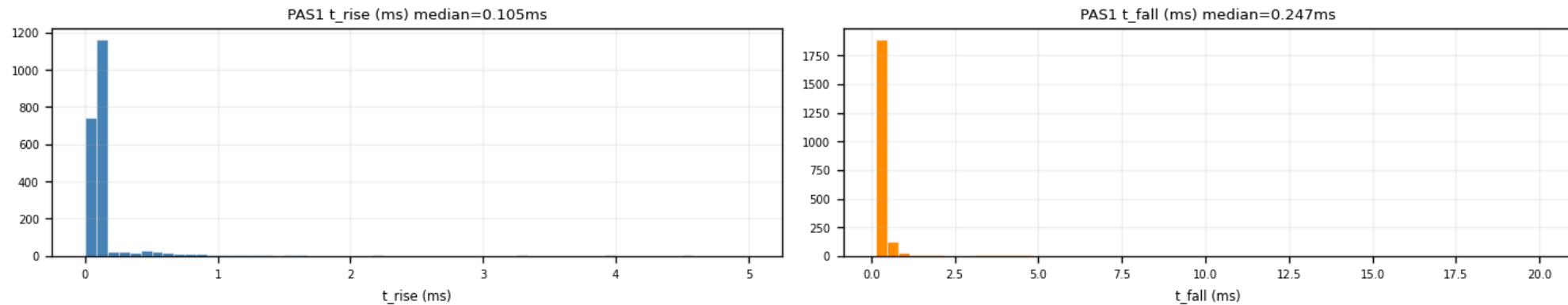
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after $\text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$ ms — final shape family (760 fits)

Backup — the $\tau_{\text{rise}} \leq 0.3$ ms ceiling (PCA input)

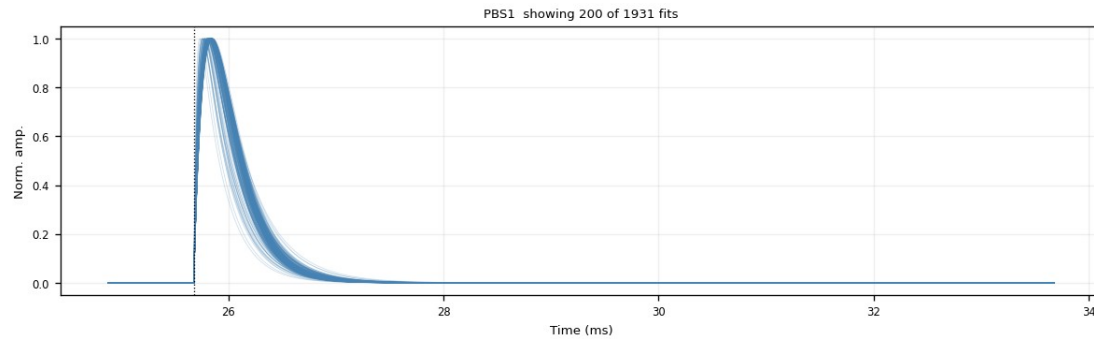
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at $\tau_{\text{rise}} \approx 0.1$ ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%) — weak 55–74% (Z22 71%) — all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



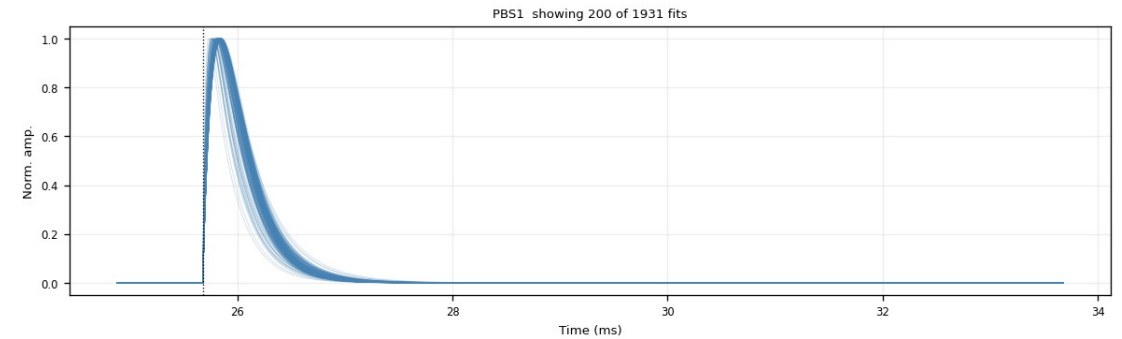
Z7 PAS1: fitted τ_{rise} (median 0.105 ms) and τ_{fall} distributions

Backup — τ_{rise} cut: a good vs a bad channel (Z7)

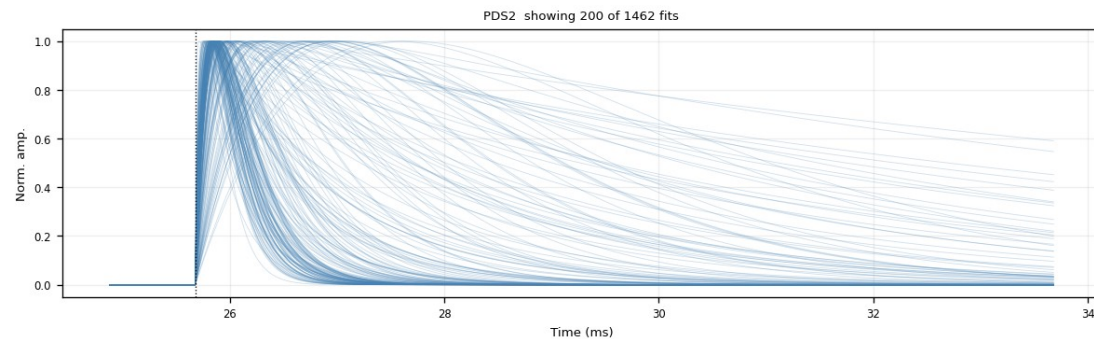
- Good channel PBS1: tight clean bundle — the $\tau_{\text{rise}} \leq 0.3$ ms cut removes $\sim 0\%$ (nearly free on quiet channels)
- Bad channel PDS2: broad, messy fan — the channel itself is bad; the cut trims the slow-rise curves, 1462 $\rightarrow$ 1154 (21%)



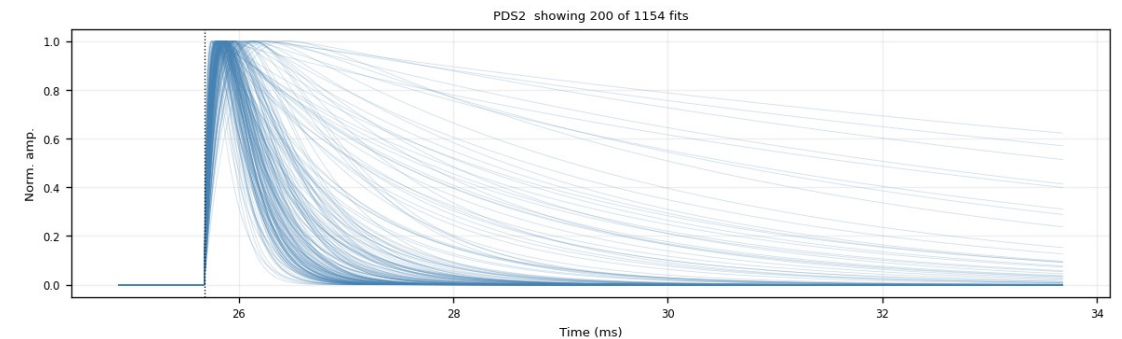
PBS1 (good) — $\text{NRMSE} \leq 0.4$



PBS1 (good) — + $\tau_{\text{rise}} \leq 0.3$ ms (unchanged)



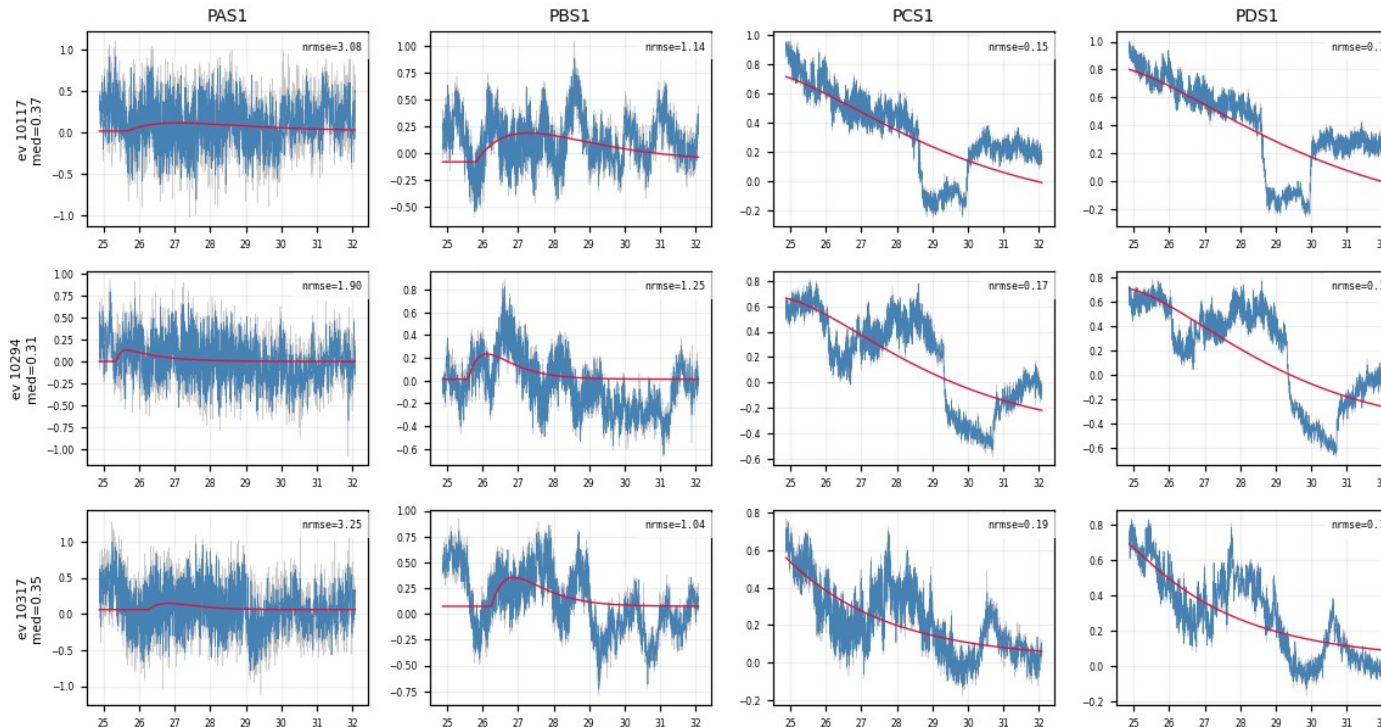
PDS2 (bad) — $\text{NRMSE} \leq 0.4$



PDS2 (bad) — + $\tau_{\text{rise}} \leq 0.3$ ms (slow risers trimmed)

Backup — what the τ_{rise} cut removes: noise NRMSE missed (Z22)

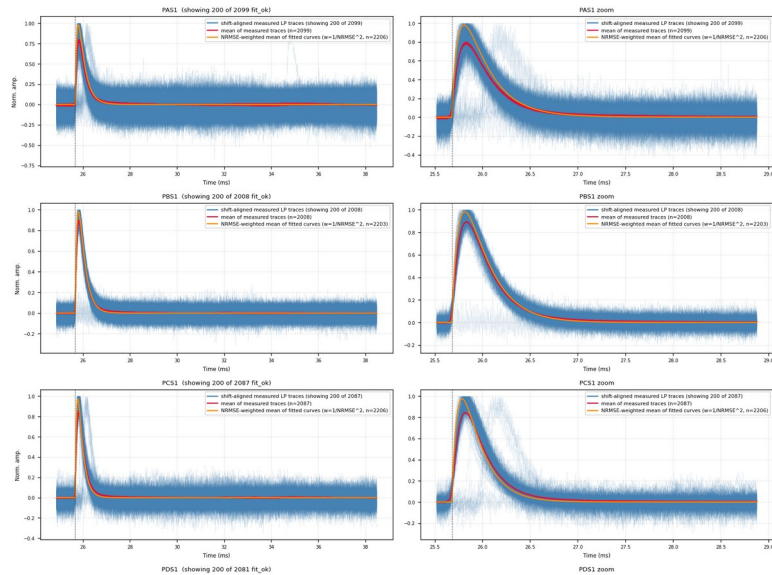
- Exactly the events that PASS $\text{NRMSE} \leq 0.4$ but are REMOVED by $\tau_{\text{rise}} \leq 0.3$ ms — the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at $\text{NRMSE} \approx 0.15$ (a slow 2-exp hugs a slow baseline drift) — but the raw trace has no clean fast pulse
- Trade-off: the same cut also trims genuine slow-rise pulses — documented, decision pending



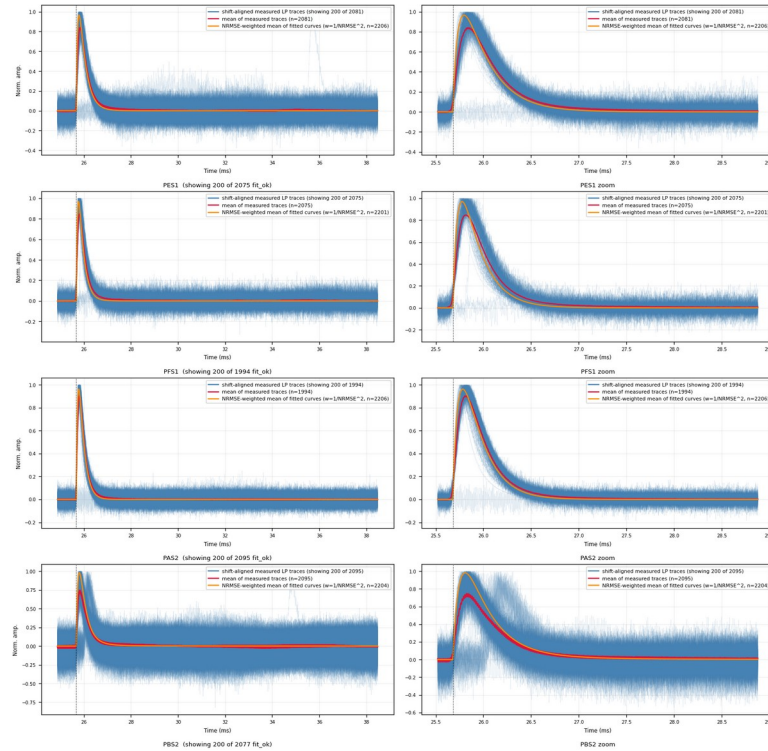
Z22: 3 events $\times$ PAS1/PBS1/PCS1/PDS1 — pass $\text{NRMSE} \leq 0.4$, removed by $\tau_{\text{rise}} \leq 0.3$ ms; PCS1/PDS1 are pure slow drift

shift-aligned measured traces · figure: zip7_lp_aligned_overlay.png

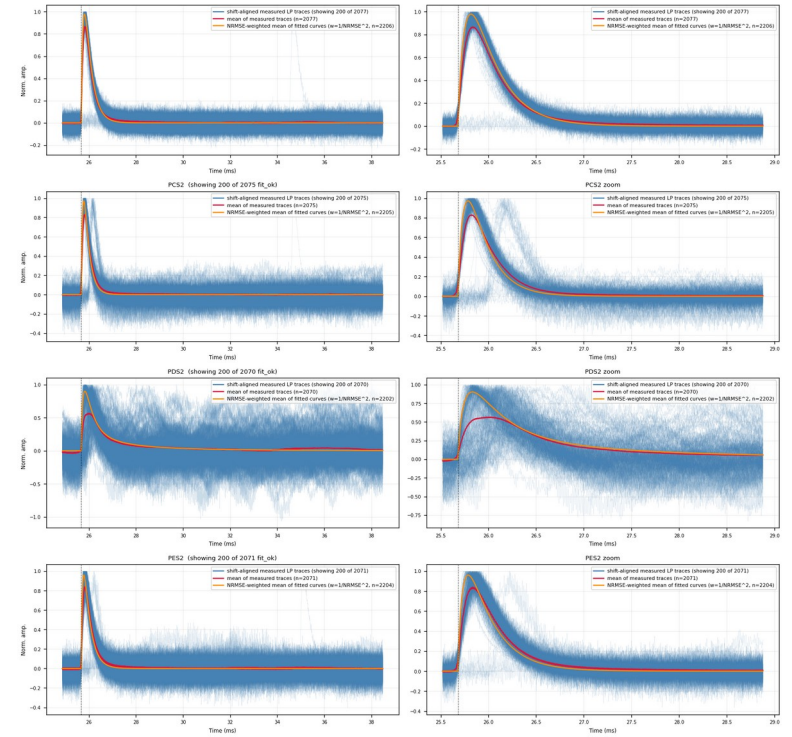
Blue = measured LP traces shifted by (fitted pretrigger - 16050); red = point-by-point mean of ALL fit_ok events; orange = NRMSE-weighted mean of the fitted curves ($w = 1/\max(\text{NRMSE}, 0.01)^2$). Rise edges line up at 16050.

[illegible]

part 1/3 (top to bottom)



part 2/3 (top to bottom)



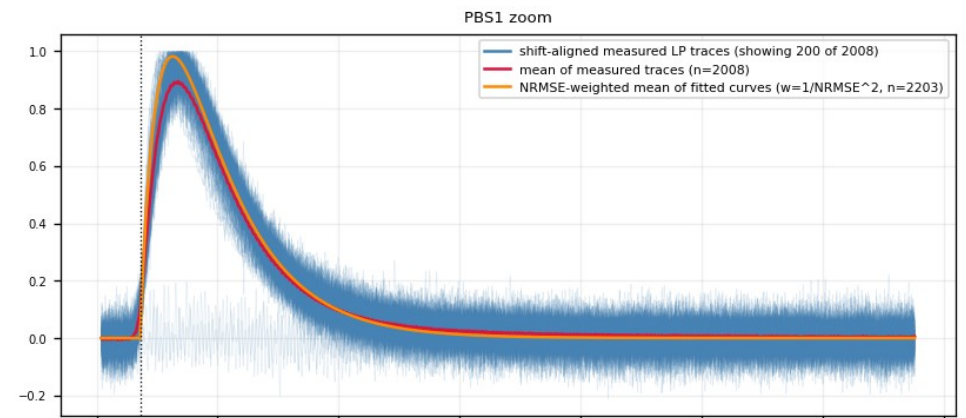
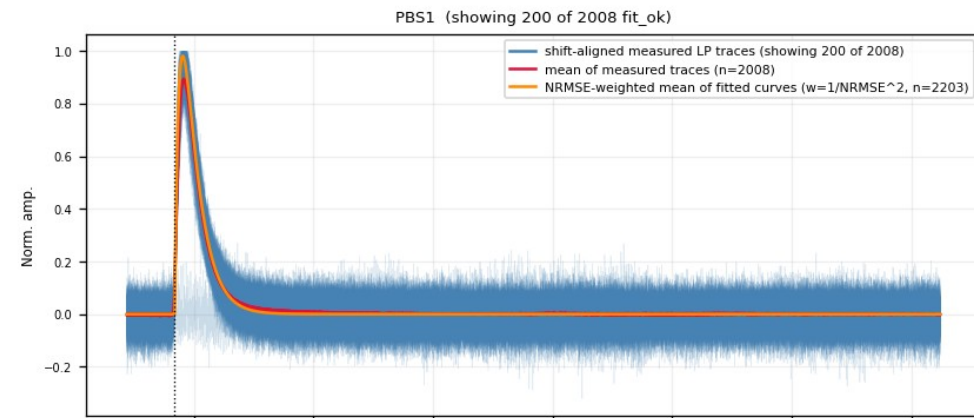
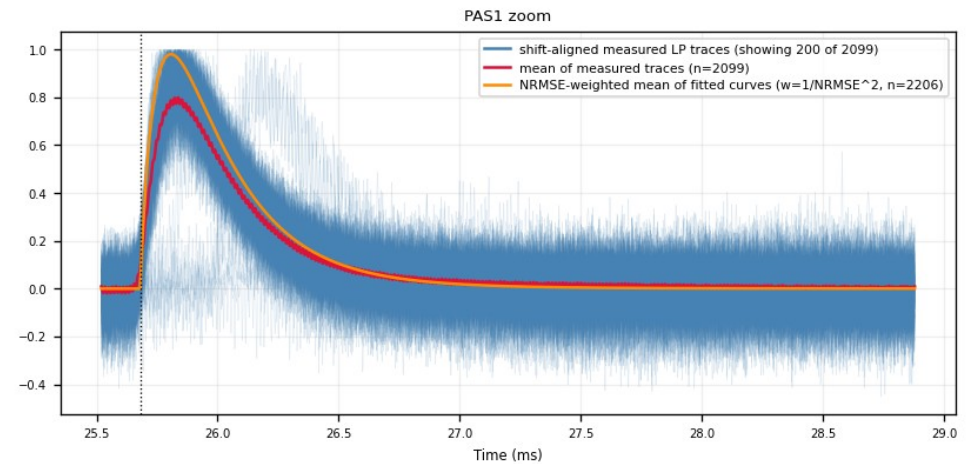
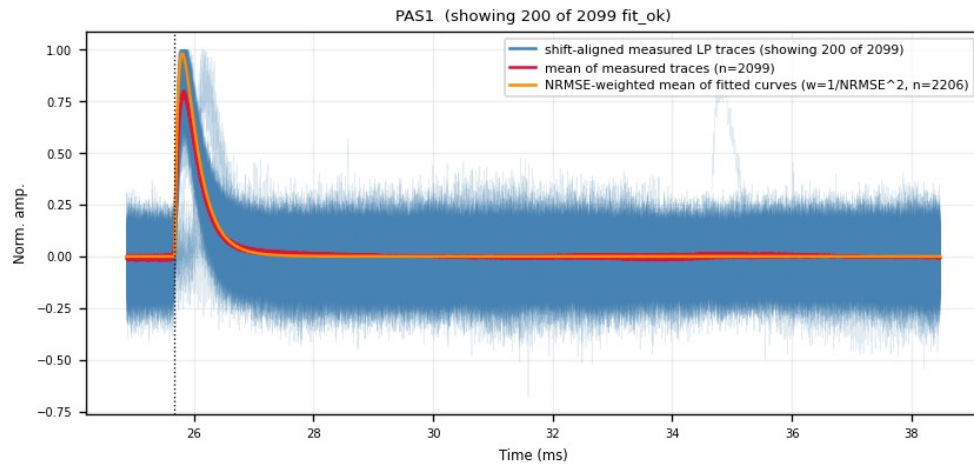
part 3/3 (top to bottom)

Backup — Z7 · aligned_overlay (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t rise<t fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: blue = shift-aligned MEASURED low-pass traces (display sample, up to 200/channel); red = point-by-point arithmetic
mean of ALL N fit_ok shifted measured traces (fit contributes only the per-event shift, no fitted curve enters this mean);
orange = NRMSE-WEIGHTED MEAN of the fitted 2-exp curves (each fit_ok event's fitted curve at common pretrigger 16050, peak-
normalized, averaged with weight  $w=1/\max(\text{NRMSE}, 0.01)^2$  so badly-fit events count less but are never excluded); dotted line =
alignment reference 16050; fit_ok only, no NRMSE cut anywhere
```

Zip7 — 100kHz LP, shift-aligned measured traces (fit_ok only, no NRMSE cut)

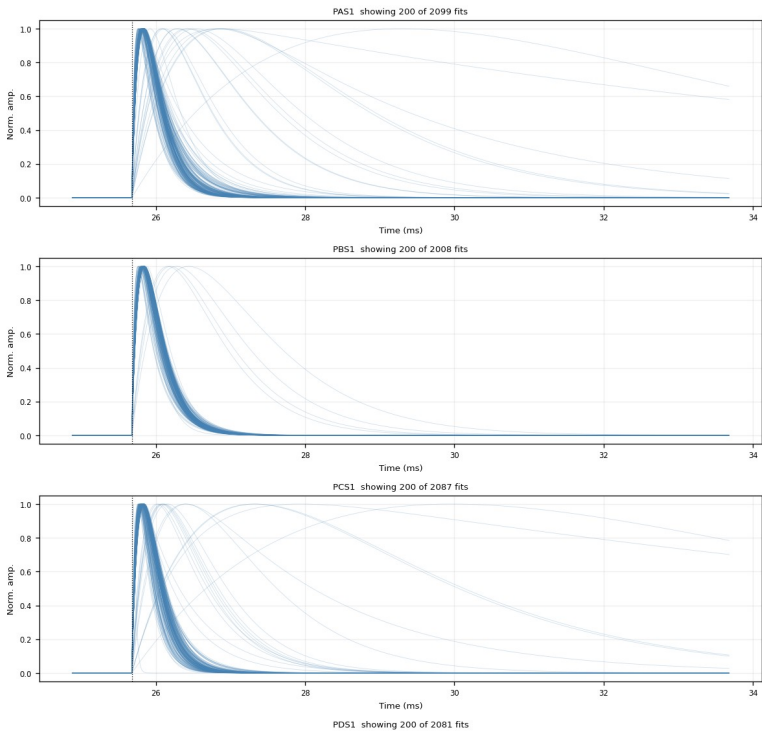


Backup — Z7 · fitted_curves_overlay

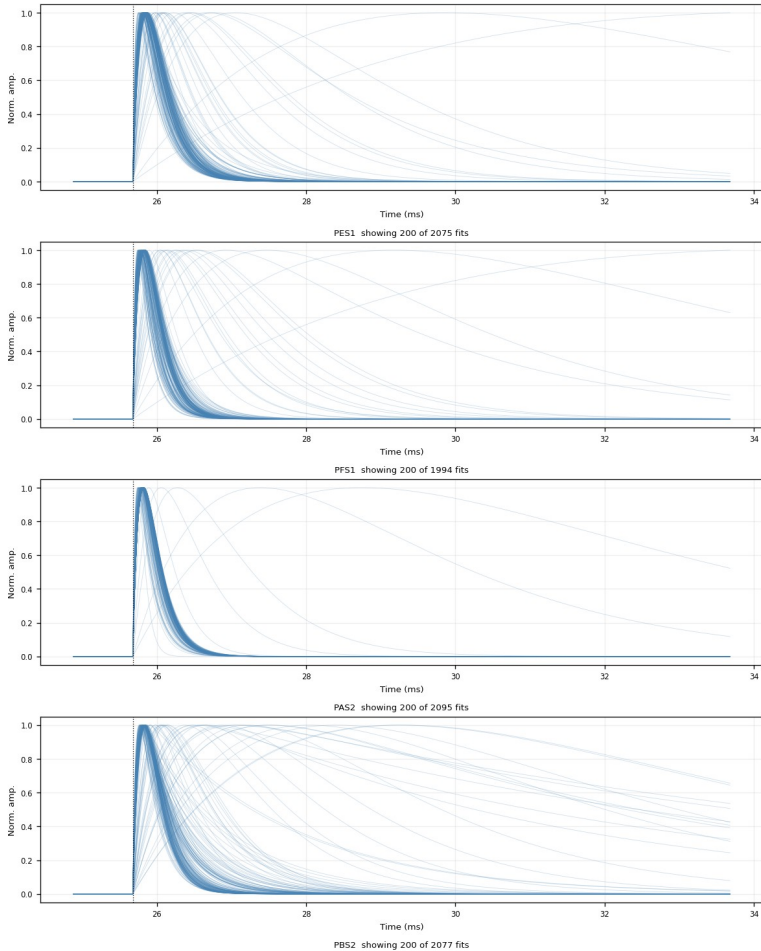
fan of fitted 2-exp curves (no cut) · figure: zip7_fitted_curves_overlay.png

Every fit_ok event's fitted curve re-evaluated at common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

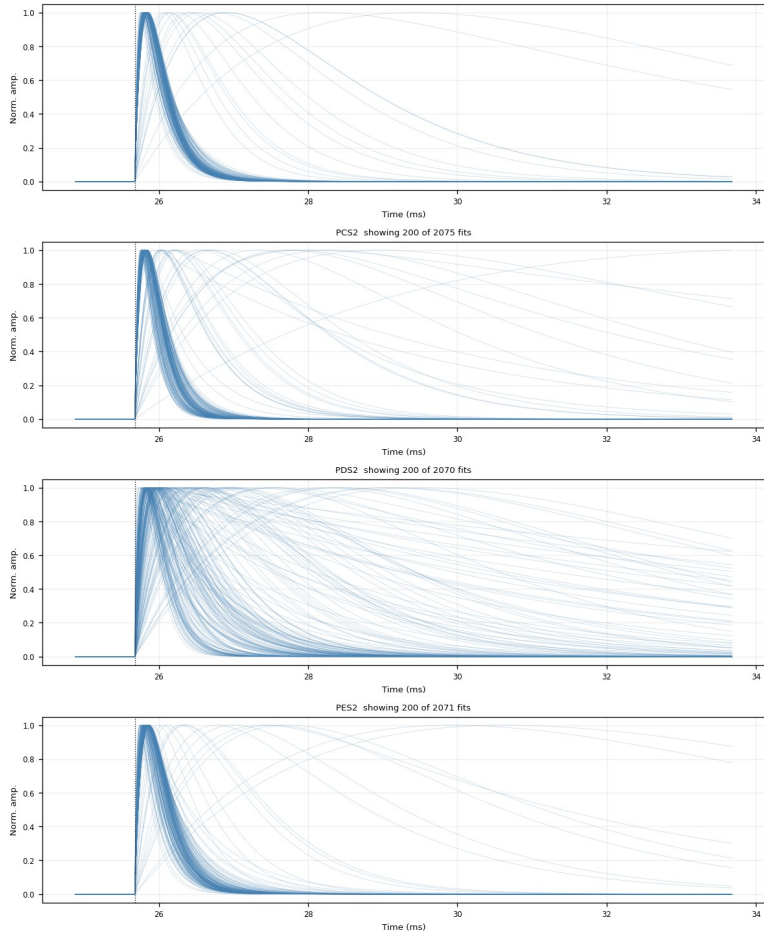
```
processing: raw MIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit (p = A*(exp(-t/tau1)-exp(-t/tau2))) -> ALL 5 param free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxrev=10000)
-> fit ok = amp and dot (rise-fall) > NRMSE < 0.001 (residuals/fitting pulse peak, RECORDED ONLY (no cut))
-> ALIGN -> shift the RECORDED trace by (fitting pretrigger - 16050) sub-sample no-interp
this figure: 9000th fitted 2-exp curves only (no measured data); each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at
common pretrigger=16050, peak-normalized, cut: fit_ok, no NRMSE cut
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, no NRMSE cut)
```



part 1/3 (top to bottom)



part 2/3 (top to bottom)



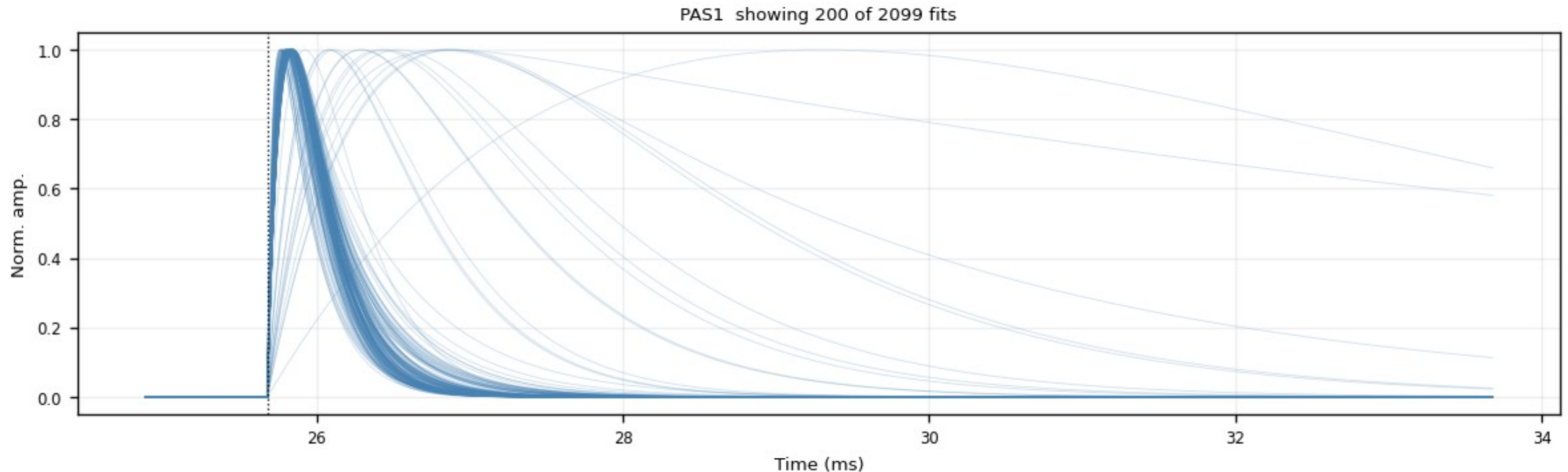
part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at
COMMON pretrigger=16050, peak-normalized; cut: fit_ok, no NRMSE cut
```

Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, no NRMSE cut)

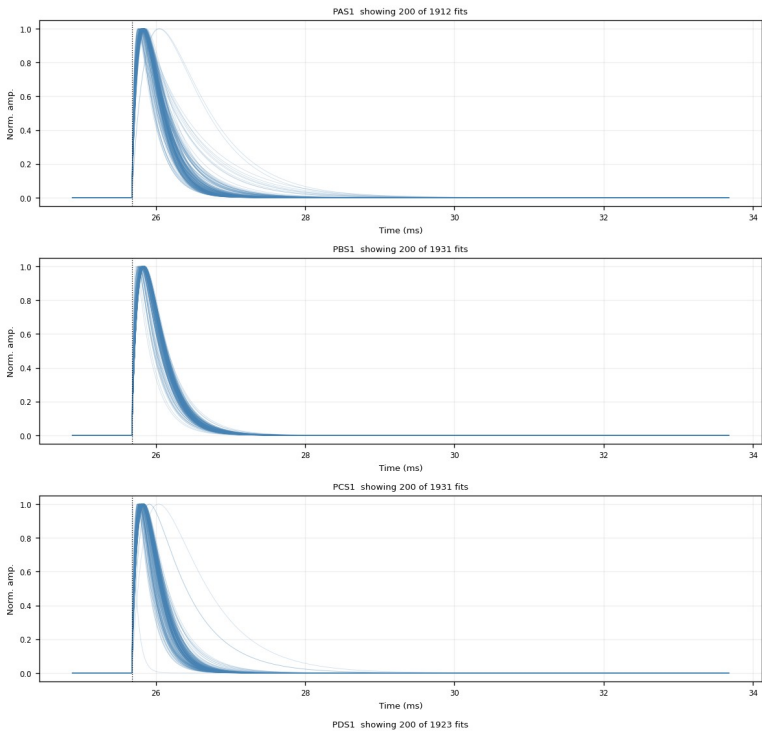


Backup — Z7 · fitted_curves_overlay

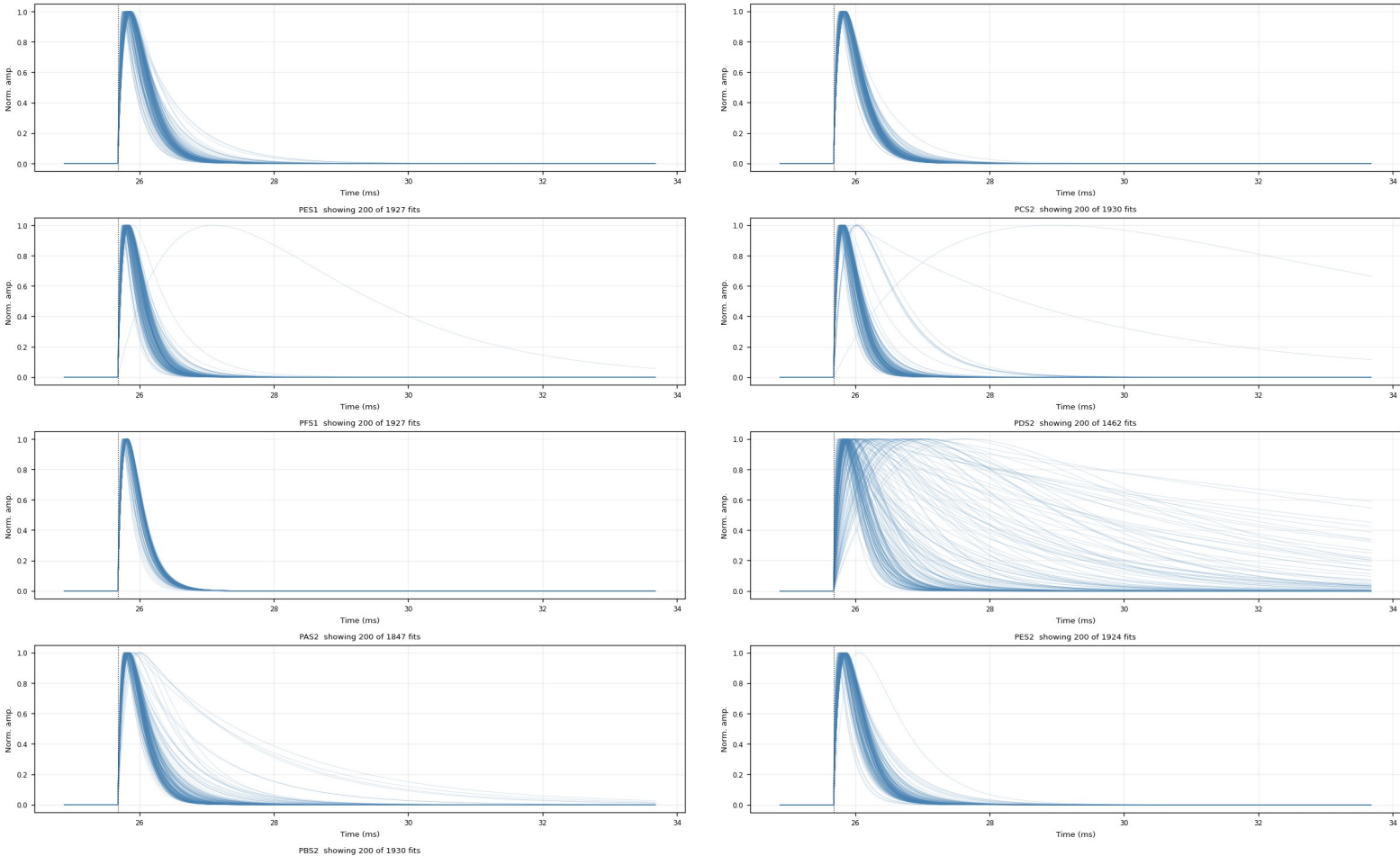
after $\text{NRMSE} \leq 0.4$ · figure: zip7_fitted_curves_overlay_nrmse0.4.png

Same fan after the NRMSE cut: the noise fan collapses; the fast-pulse bundle remains.

processing: raw MIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/\tau_1) + \exp(-t/\tau_2))$; AL5 5 param free incl. pretrigger (curve fit; pretrigger in 16050-3000, maxrev=10000)
-> fit ok = amp and dot (rise-fall) > NRMSE < 0.4; NRMSE = RMS(fit residual)/fitted pulse peak; RECORDED ONLY (no cut)
Align -> shift the RECORDED traces by (fitted pretrigger - 16050) sub-sample 90-integ
this figure: 90000 fitted 2-exp curves only (no measured data); each fit ok event's fitted (amp, τ_1 , τ_2) re-evaluated at
common pretrigger=16050; peak-normalized; cut: fit_ok, NRMSE<=0.4
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4)



part 1/3 (top to bottom)



part 2/3 (top to bottom)

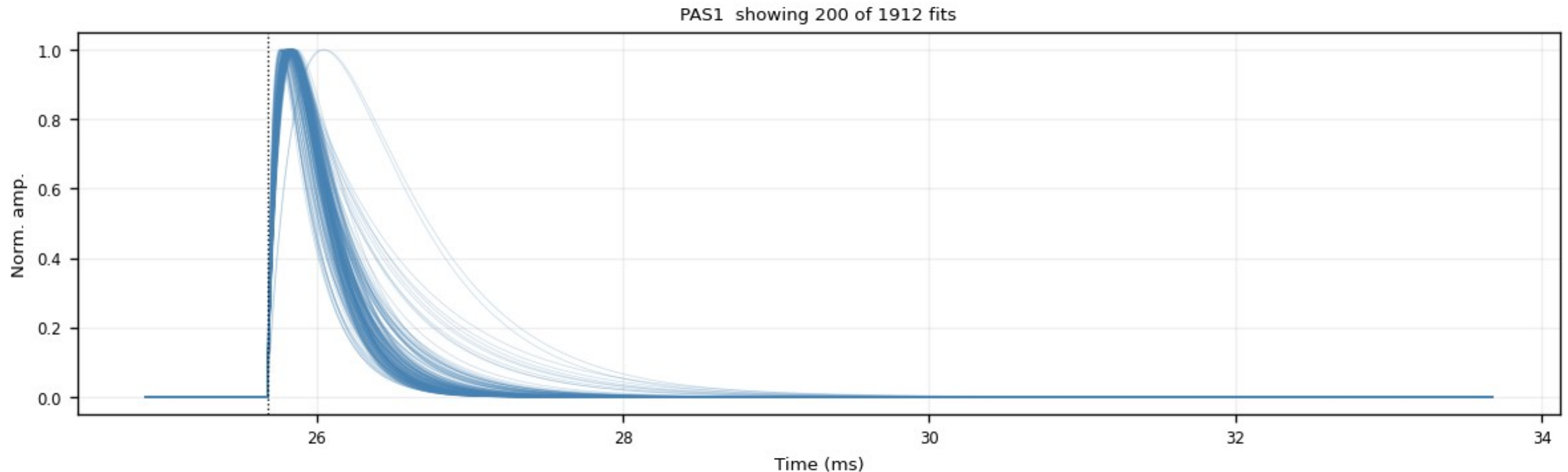
part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

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processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at
COMMON pretrigger=16050, peak-normalized; cut: fit_ok, NRMSE<=0.4
```

Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4)



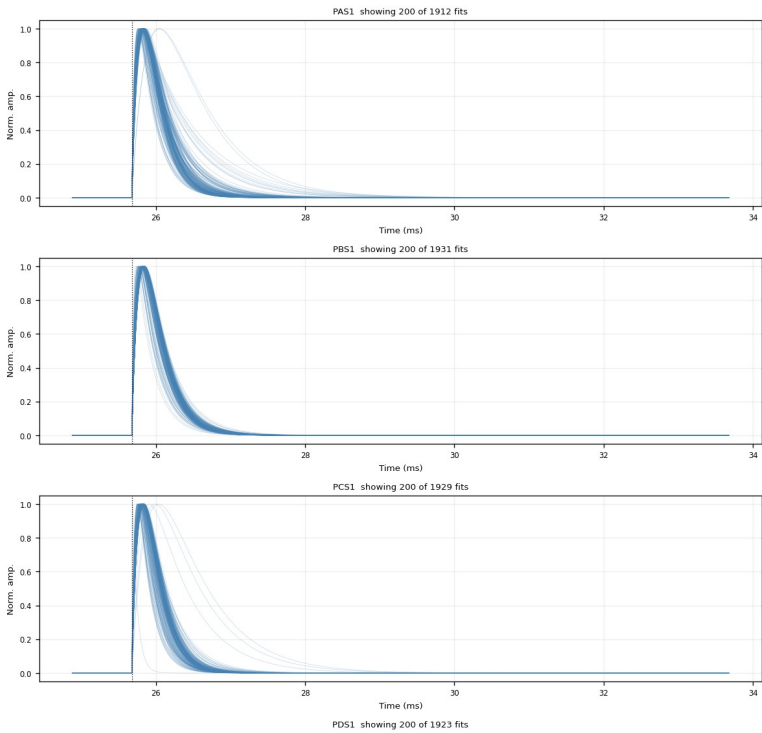
Backup — Z7 · fitted_curves_overlay

after $\text{NRMSE} \leq 0.4$ and $\tau_{\text{rise}} \leq 0.3 \text{ ms}$ · figure: zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png

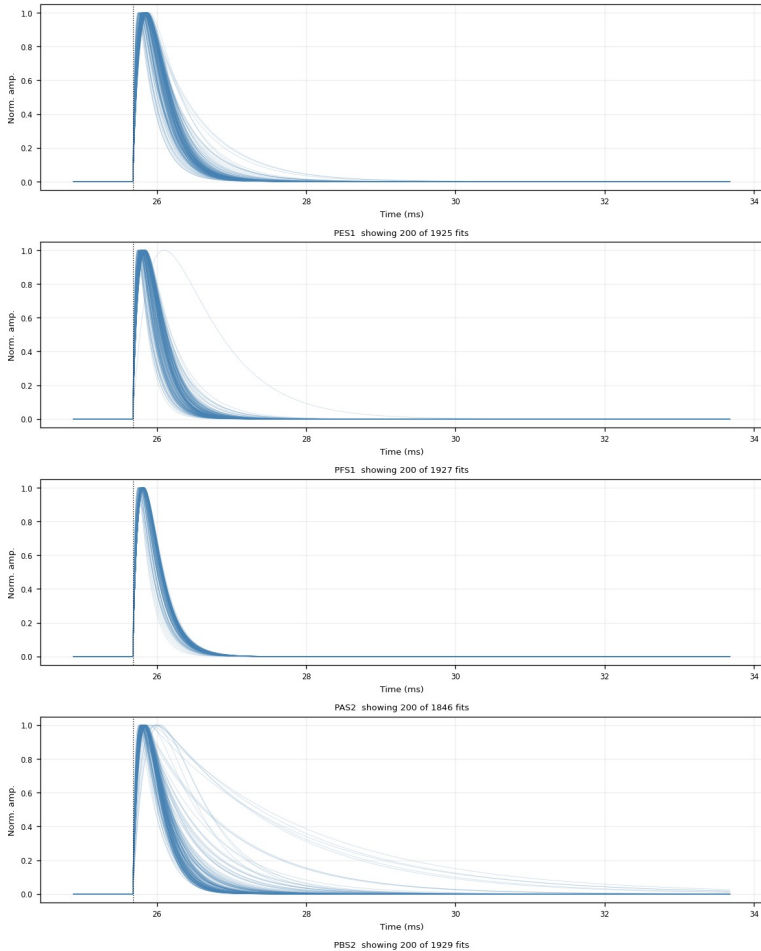
The exact PCA input population: clean fast-pulse bundle only.

processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$, AL5 5 param free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxrev=10000)
-> fit ok = amp and dot (rise+fall) > NRMSE < 0.4 (NRMSE = RMSE/fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align -> shift the RECORDED trace by (fitted pretrigger - 16050) sub-sample no-interp
this figure: 9000th fitted 2-exp curves only (no measured data), each fit ok event's fitted (amp, τ_{rise} , τ_{fall}) re-evaluated at
common pretrigger=16050, peak-normalized, cut: fit_ok, NRMSE<=0.4, $\tau_{\text{rise}}<=0.30\text{ms}$

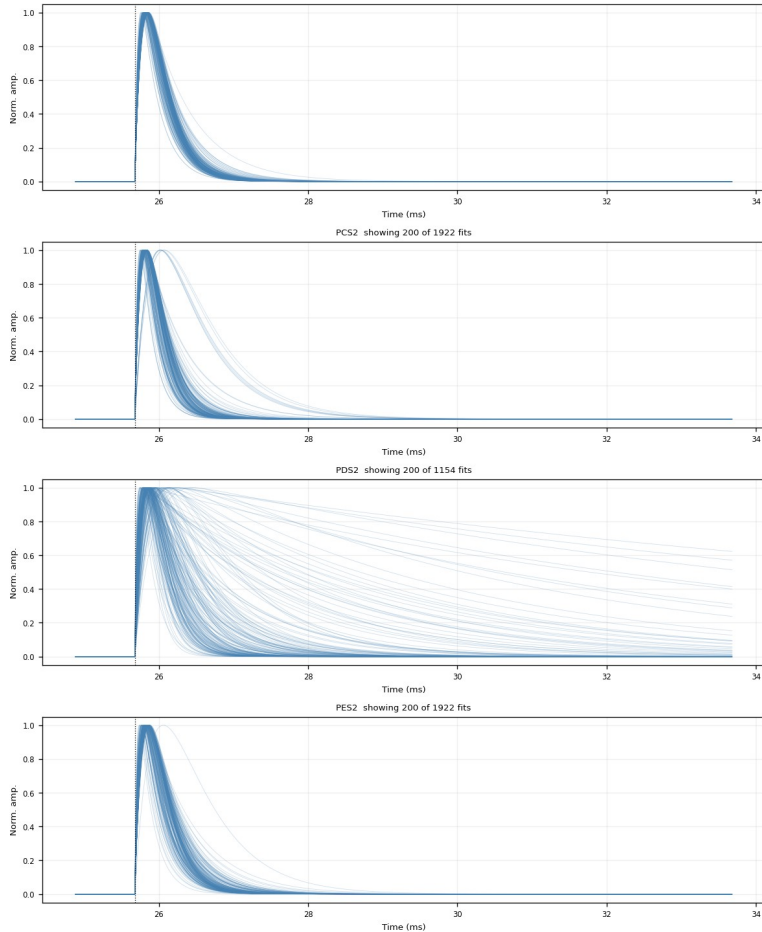
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4, $\tau_{\text{rise}}<=0.30\text{ms}$)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



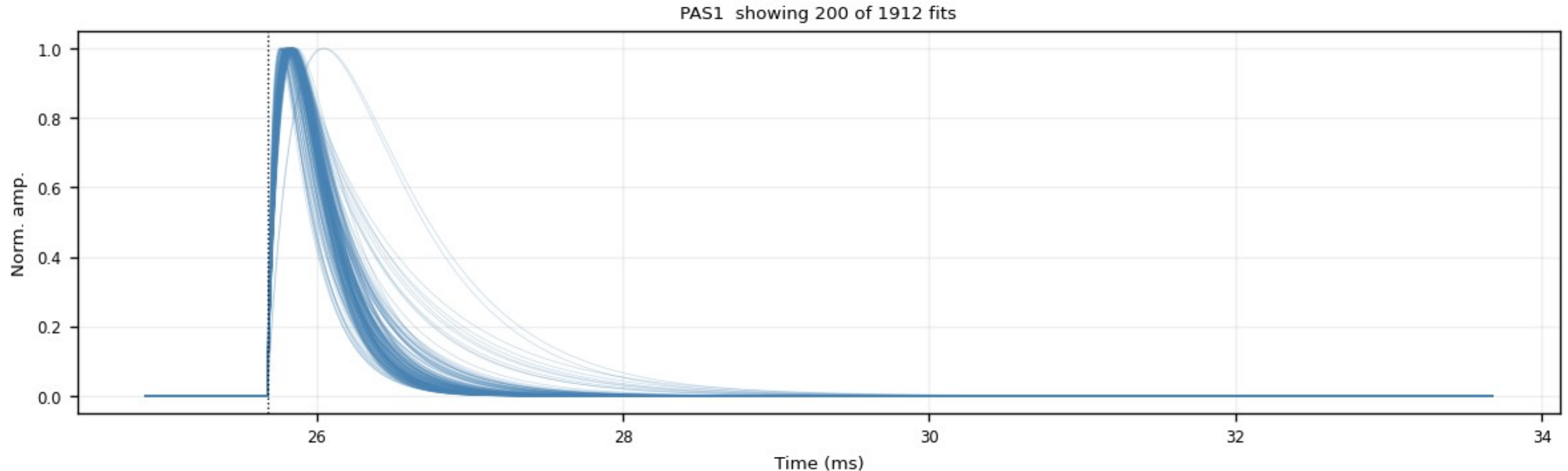
part 3/3 (top to bottom)

Backup — Z7 · fitted_curves_overlay (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at
COMMON pretrigger=16050, peak-normalized; cut: fit_ok, NRMSE<=0.4, t_rise<=0.30ms
```

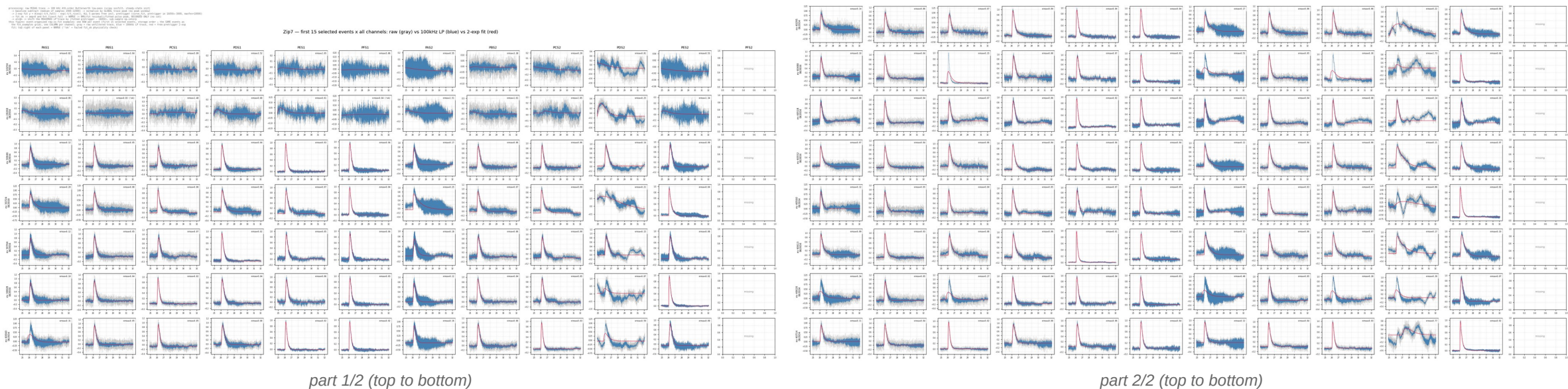
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4, t_rise<=0.30ms)



Backup — Z7 · raw_vs_fit_examples

first 15 events × all channels · figure: zip7_raw_vs_fit_examples.png

Gray = raw unfiltered, blue = 100 kHz LP, red = 2-exp fit; per-panel NRMSE top-right. Same 15 events as fit_examples; noise vs pulse size is directly visible.

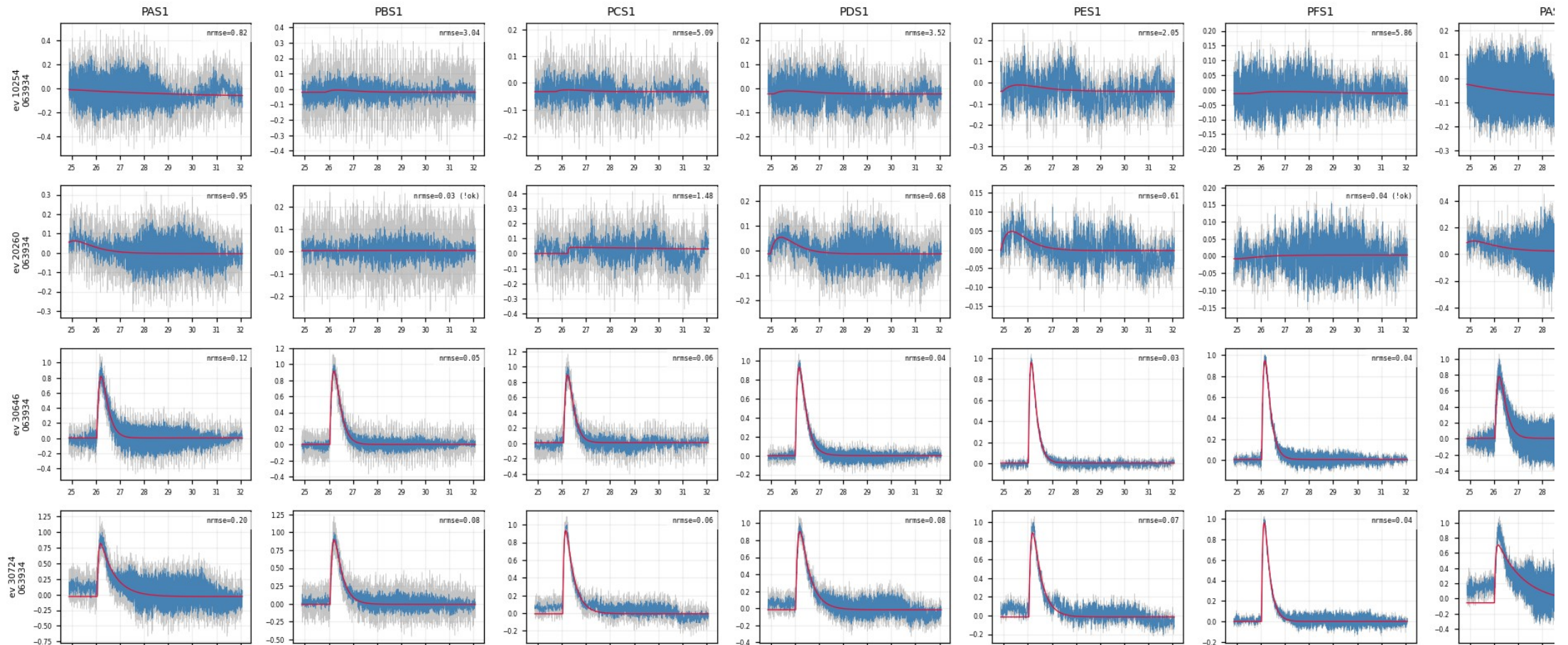


Backup — Z7 · raw_vs_fit_examples (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: event-organized raw-vs-fit examples: one ROW per event (first 15 selected events, storage order – the SAME events as
the fit_examples grid), one COLUMN per channel; gray = raw unfiltered trace, blue = 100kHz LP trace, red = free-pretrigger 2-exp
fit; top-right of each panel = NRMSE ('!ok' = failed fit_ok physicality check)
```

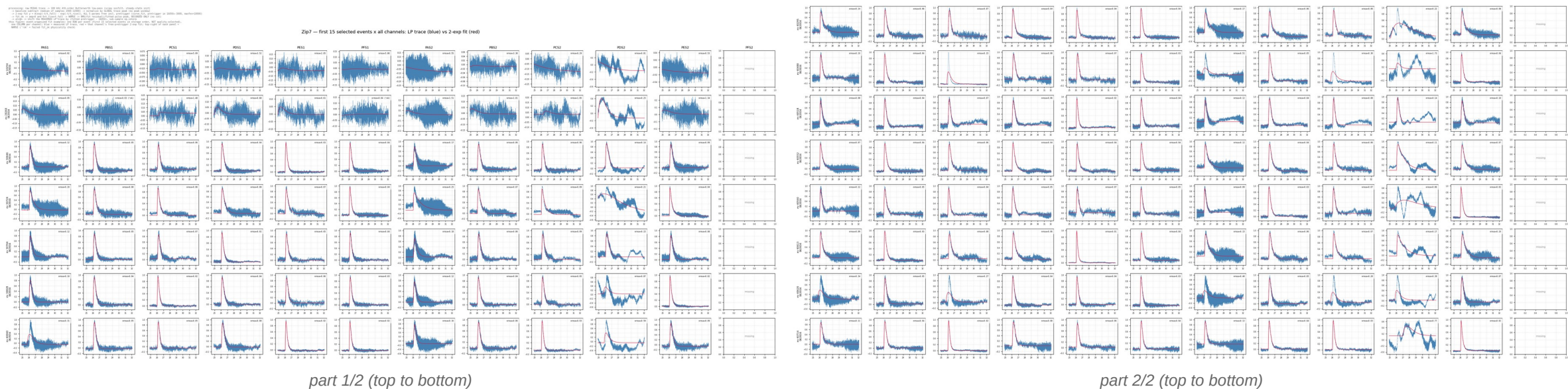
Zip7 — first 15 selected events x all channels: raw (gray) vs 100



Backup — Z7 · fit_examples

first 15 events × all channels (LP vs fit) · figure: zip7_fit_examples.png

One ROW per event, one COLUMN per channel. A real event shows a clean pulse and a good fit in every channel simultaneously; a noise trigger fails across the whole row.

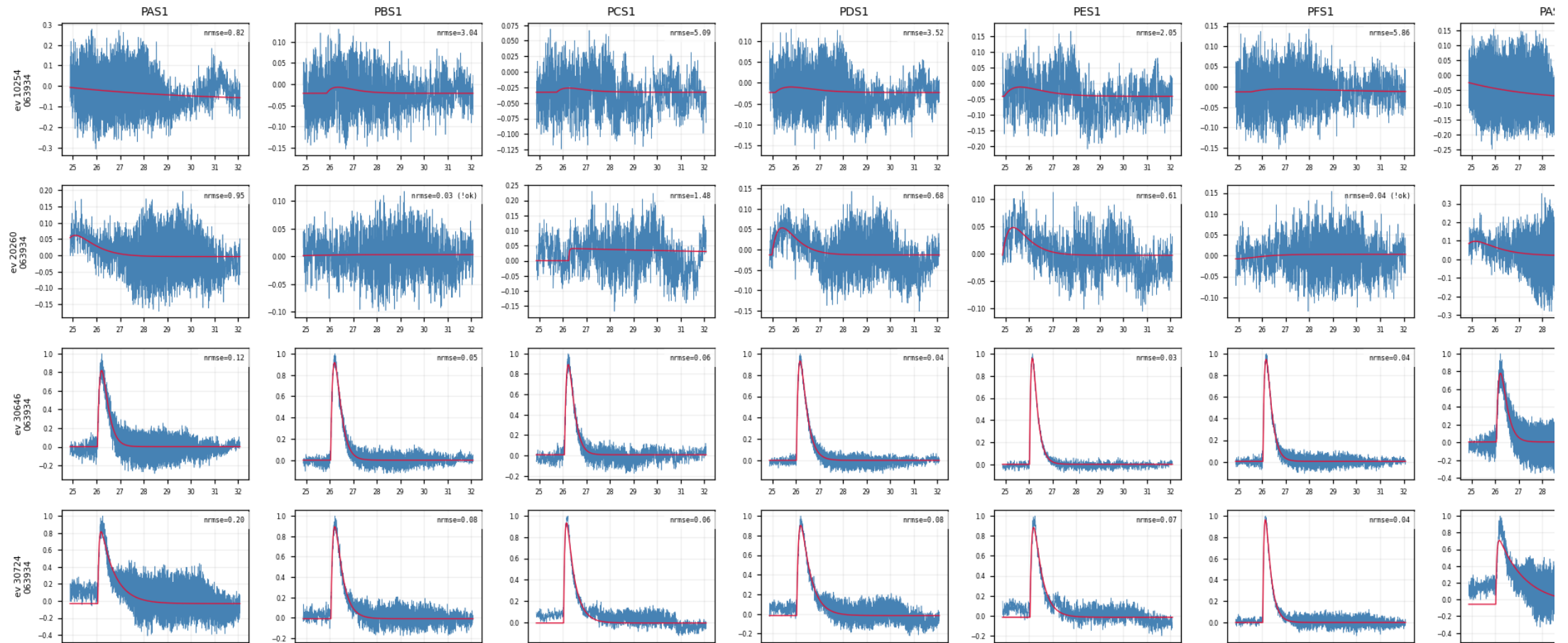


Backup — Z7 · fit_examples (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t rise<t fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: event-organized fit examples: one ROW per event (first 15 selected events in storage order, NOT quality-selected),
one COLUMN per channel; blue = measured LP trace, red = that channel's free-pretrigger 2-exp fit; top-right of each panel =
NRMSE ('!ok' = failed fit_ok physicality check)
```

Zip7 — first 15 selected events x all channels: LP trace (



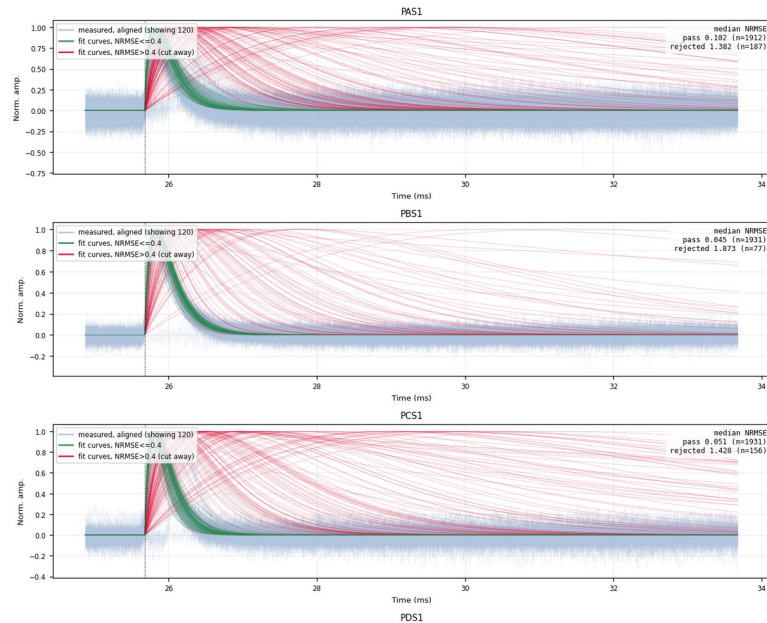
Backup — Z7 · overlay_fan_cut

measured traces + kept/rejected fits · figure: zip7_overlay_fan_cut_nrmse0.4.png

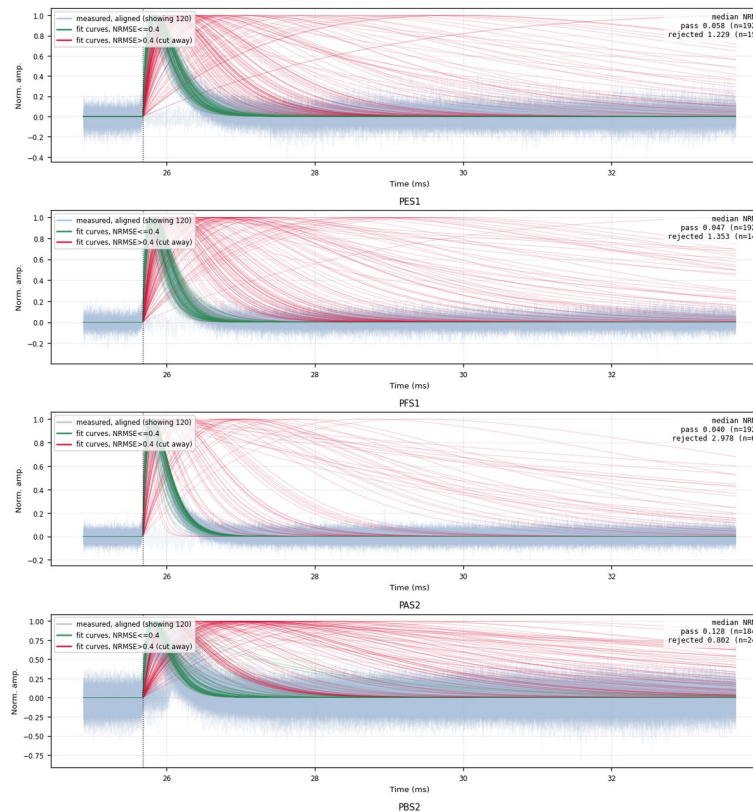
Gray-blue = aligned measured traces; green = fitted curves with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations stamped per channel.

```
processing: raw NIDAC trace -> 100 kHz 4th-order Butterworth low-pass (slip offset, steady-state lock)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-emp fit v = amp0 + v1 * t + v2 * t^2 + v3 * t^3 + v4 * t^4 + v5 * t^5; pretrigger (curve fit, pretrigger in 16050-3000, maxv=10000)
-> fit ok -> amp0 and get t1, t2, t3, t4, t5 -> NRMSE -> RMS fit residual/fitted pulse peak; RECORDED ONLY (no cut)
-> align -> shift the RECORDED LP trace by fitted pretrigger - 16050; non-linear no-align
this figure: three views in one panel: short-aligned RECORDED LP traces (gray-blue; fit ok, 10 mean lines) + SMOOTH fitted 2-emp
curves at common pretrigger 16050; peak normalized; split by the NRMSE cut: green = NRMSE <= 0.4 (kept); red = NRMSE > 0.4 (cut
away); top-right box = median NRMSE and counts of each population
```

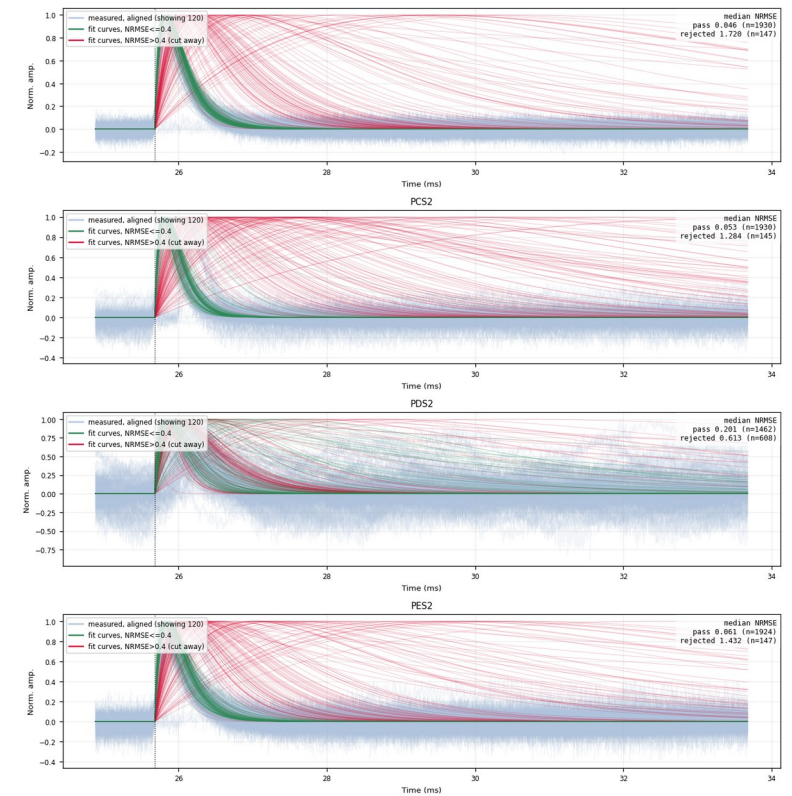
Zip7 — aligned measured traces + fitted curves split by NRMSE cut 0.4



part 1/3 (top to bottom)



part 2/3 (top to bottom)



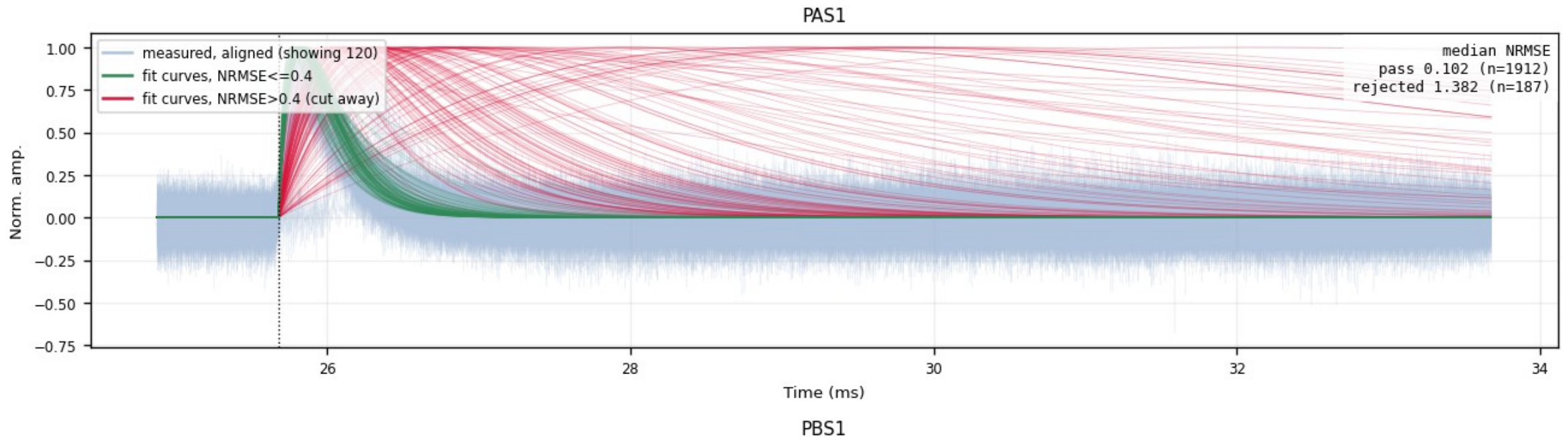
part 3/3 (top to bottom)

Backup — Z7 · overlay_fan_cut (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: three views in one panel: shift-aligned MEASURED LP traces (gray-blue, fit ok, NO mean line) + SMOOTH fitted 2-exp
curves at common pretrigger 16050, peak-normalized, split by the NRMSE cut: green = NRMSE<=0.4 (kept), red = NRMSE>0.4 (cut
away); top-right box = median NRMSE and counts of each population
```

Zip7 — aligned measured traces + fitted curves split by NRMSE cut 0.4



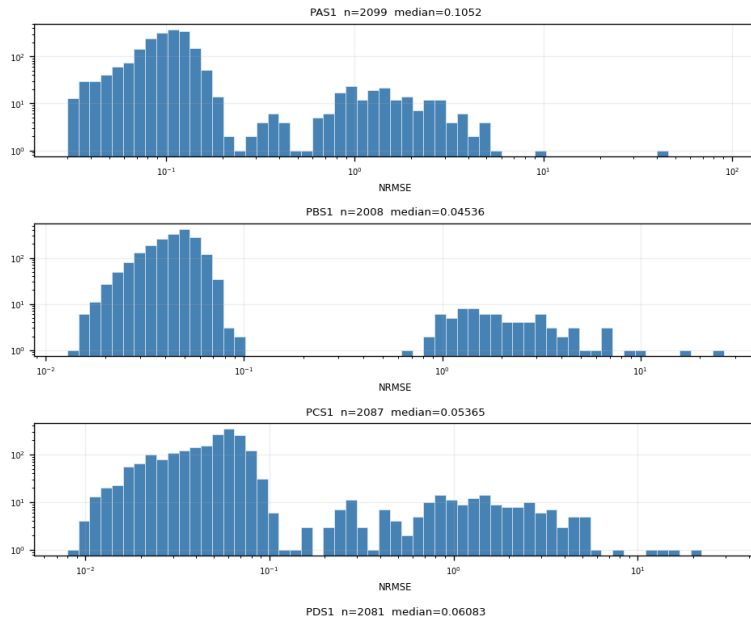
Backup — Z7 · nrmse

NRMSE distribution (log-log) · figure: zip7_nrmse.png

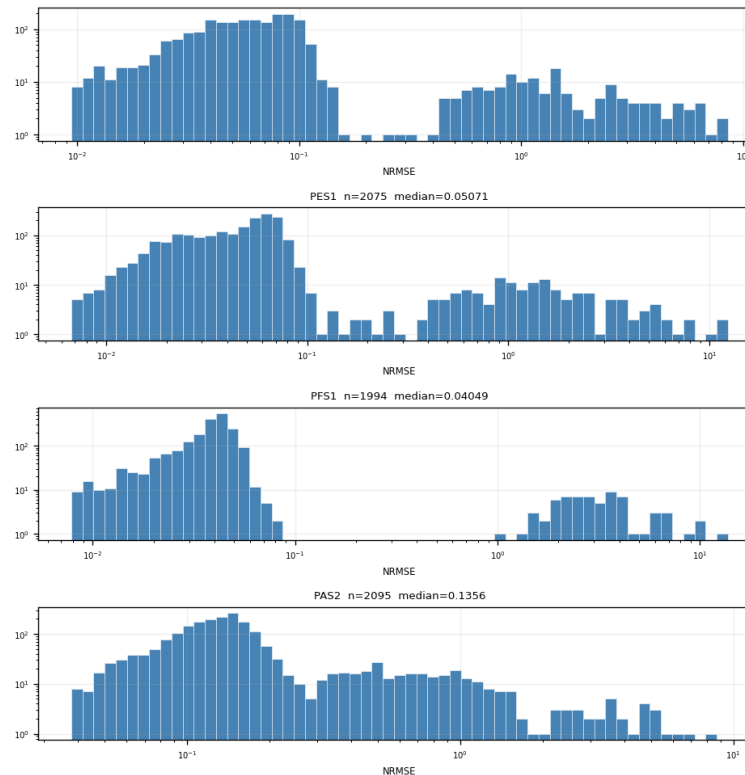
NRMSE = RMS(fit residual)/fitted peak. Bimodal: good-fit population ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets the cut.

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxfev=10000)
-> fit ok := amp0 and 0 < rise < fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NRMSE distribution of fit ok events, log-log axes (log-spaced bins, log counts); binodal = good-fit population vs
noise triggers; valley = natural cut candidate; NO cut applied
```

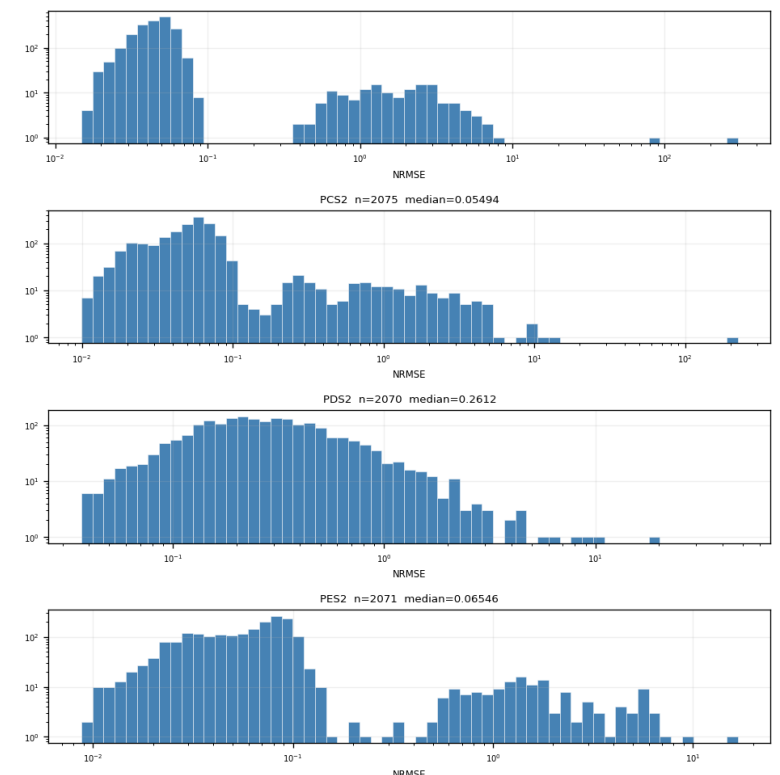
Zip7 — nrmse distribution (fit_ok events)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



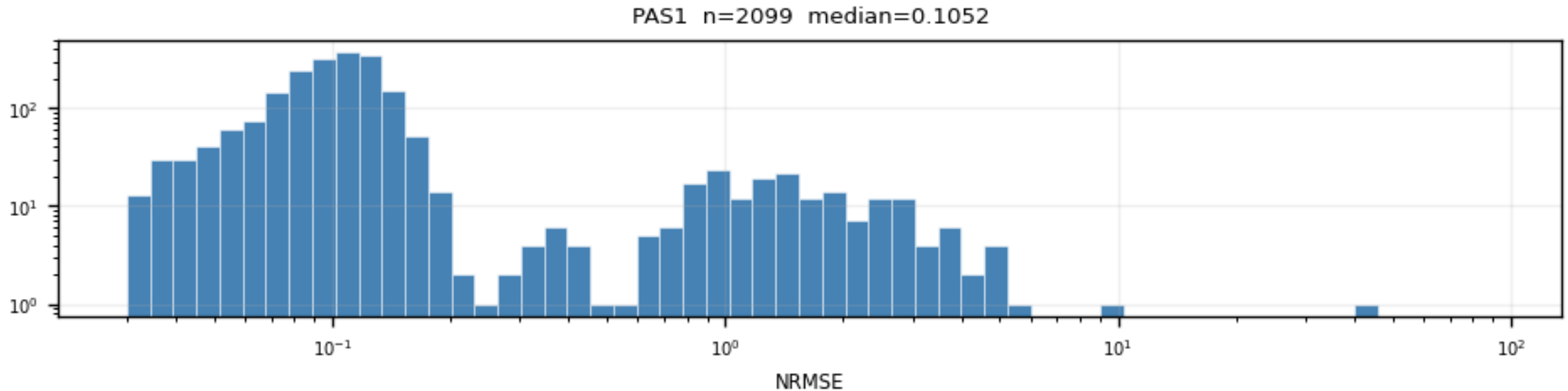
part 3/3 (top to bottom)

Backup — Z7 · nrmse (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NRMSE distribution of fit_ok events, log-log axes (log-spaced bins, log counts); bimodal = good-fit population vs
noise triggers; valley = natural cut candidate; NO cut applied
```

Zip7 — nrmse distribution (fit_ok events)



PBS1 n=2008 median=0.04536

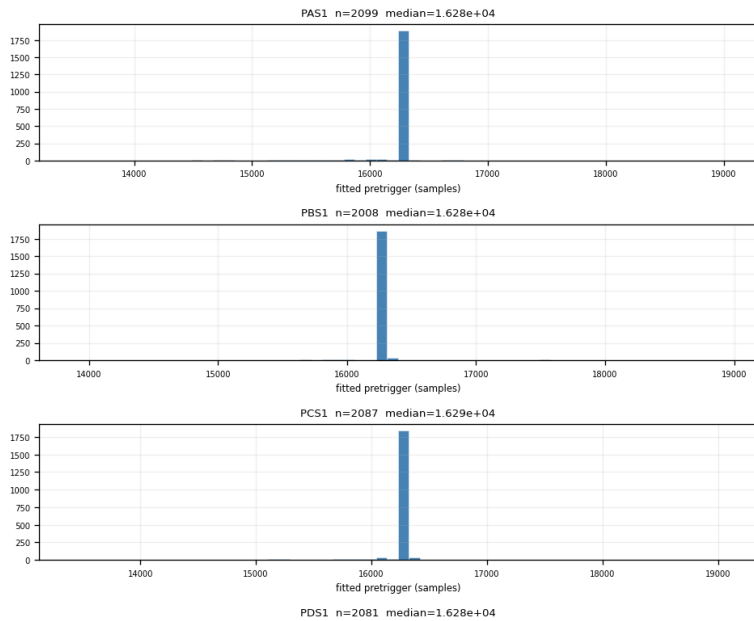
Backup — Z7 · pretrigger

fitted onset distribution · figure: zip7_pretrigger.png

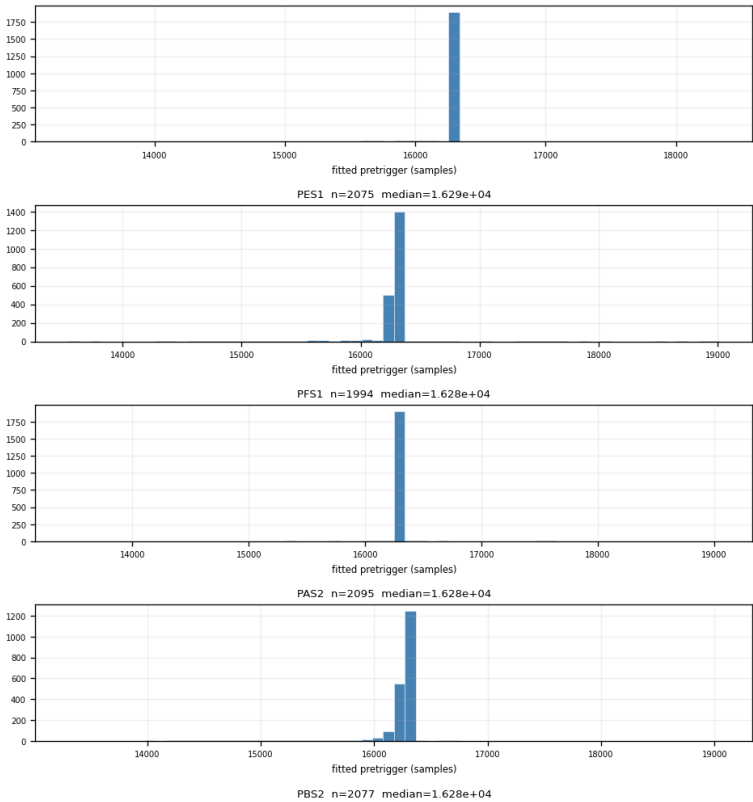
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050, which is why pinning it inside the fit was wrong.

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxfev=10000)
-> fit ok := amp>0 and 0<rise<fall -> NPWE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050); sub-sample np.interp
this figure: distribution of the FITTED free pretrigger (samples) for fit_ok events; reference 16050

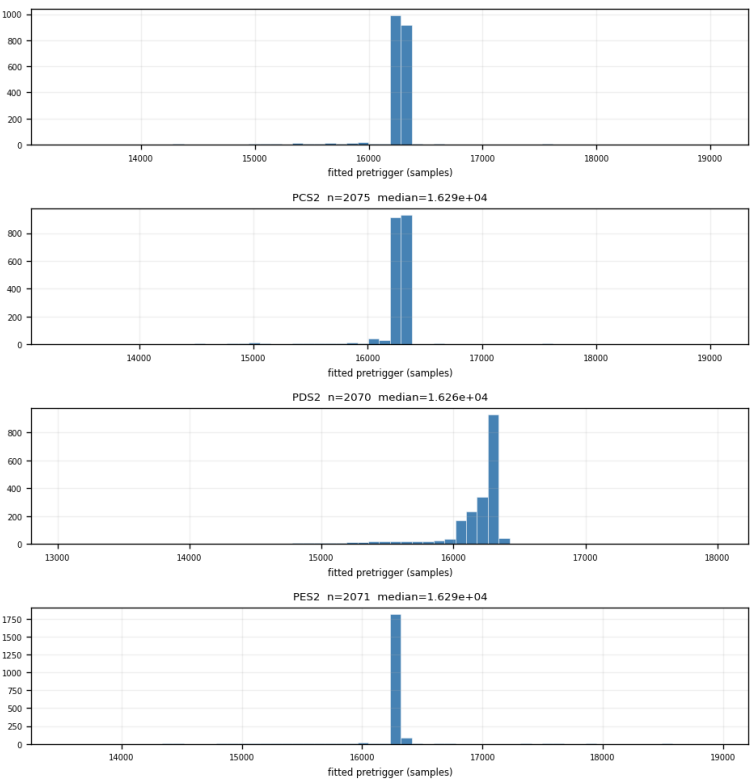
Zip7 — pretrigger distribution (fit_ok events)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



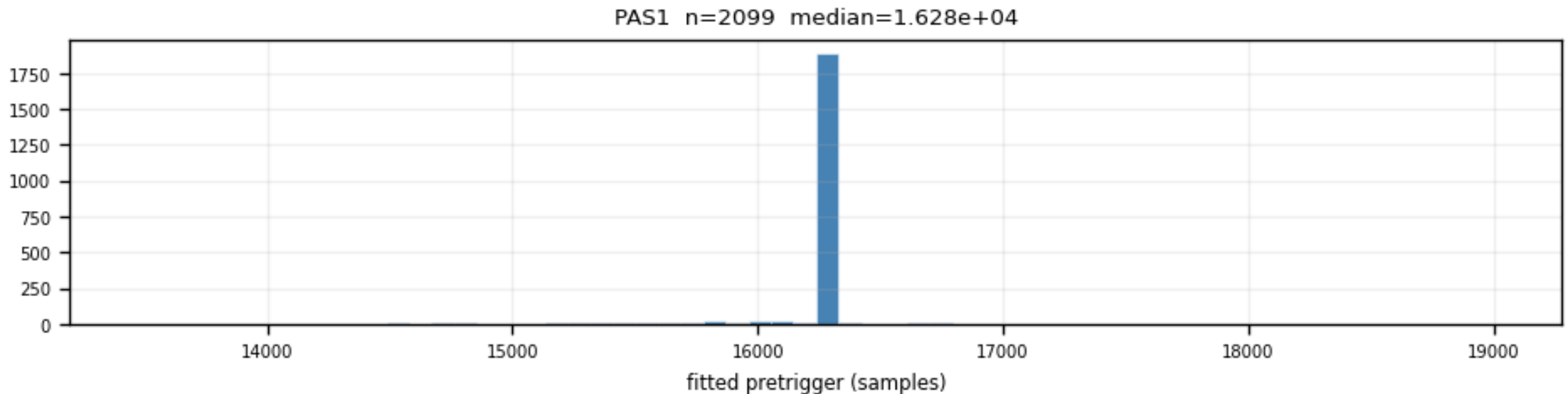
part 3/3 (top to bottom)

Backup — Z7 · pretrigger (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: distribution of the FITTED free pretrigger (samples) for fit_ok events; reference 16050
```

Zip7 — pretrigger distribution (fit_ok events)



PBS1 n=2008 median=1.628e+04

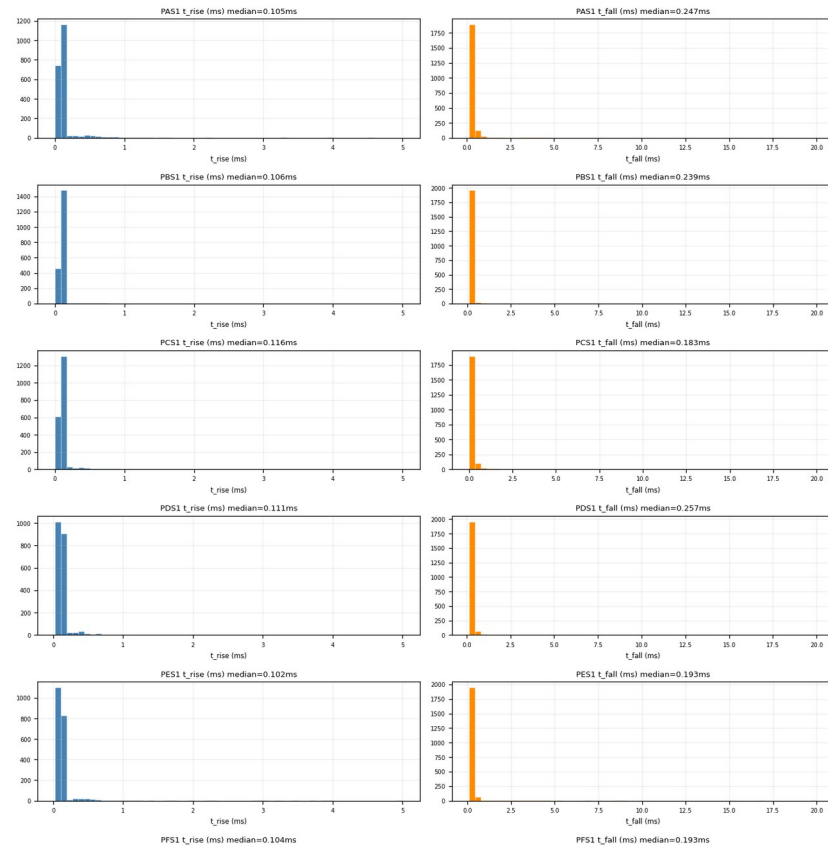
Backup — Z7 · time_constants

fitted τ_{rise} / τ_{fall} distributions · figure: zip7_time_constants.png

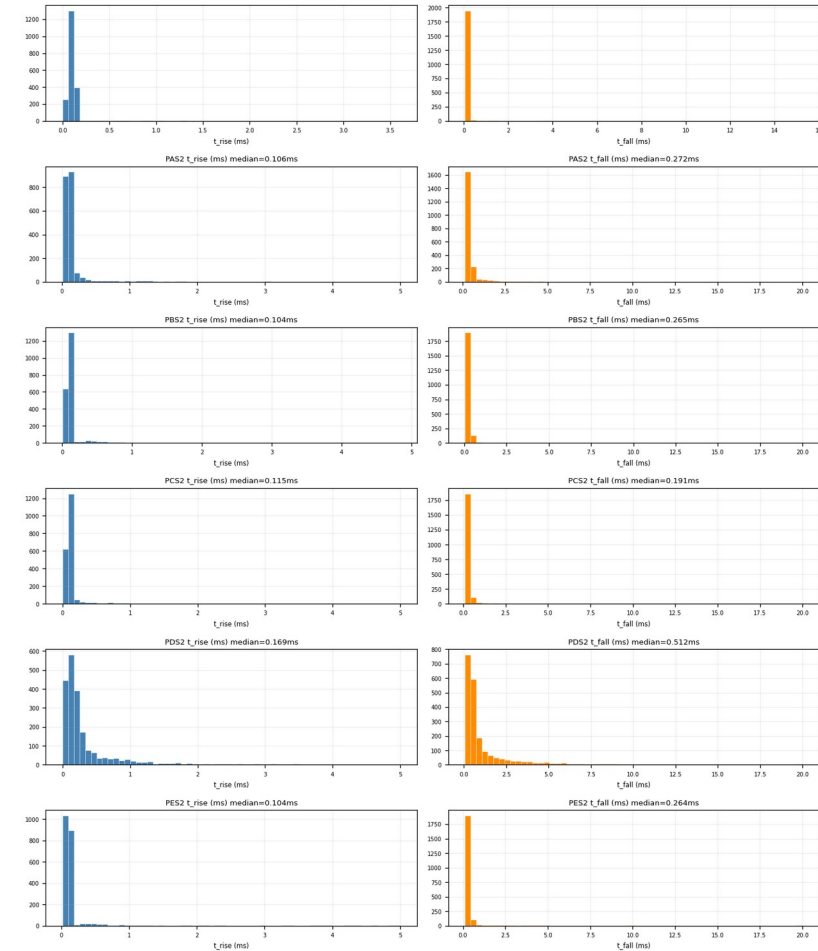
Fast-pulse population sits at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds (5 ms / 20 ms) are fits pinned at the fit limits (noise), removed by the NRMSE cut.

```
processing: raw MDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (mean of samples 2000-2200) -> normalize by 500mV trace peak (no peak clipping)
-> 2-amp fit y = A*exp(-t/tau_fall) - exp(-t/tau_rise); All 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3800, maxcov=10000)
-> fit on 10 samples and best fitted fall -> NRMSE -> 100/fits_residuals/fitted pulse peak, RECORDED ONLY (no cut)
-> align -> shift the RECORDED LP trace by (fitted pretrigger - 16050), sub-sample 10, jitter
this figure: fitted_tau_rise / tau_fall distributions (ms) of fit_ok events; free-pretrigger fit, loose bounds tau_rise=5ms tau_fall=20ms
```

Zip7 — τ_{rise} / τ_{fall} (fit_ok events, free-pretrigger fit)



part 1/2 (top to bottom)



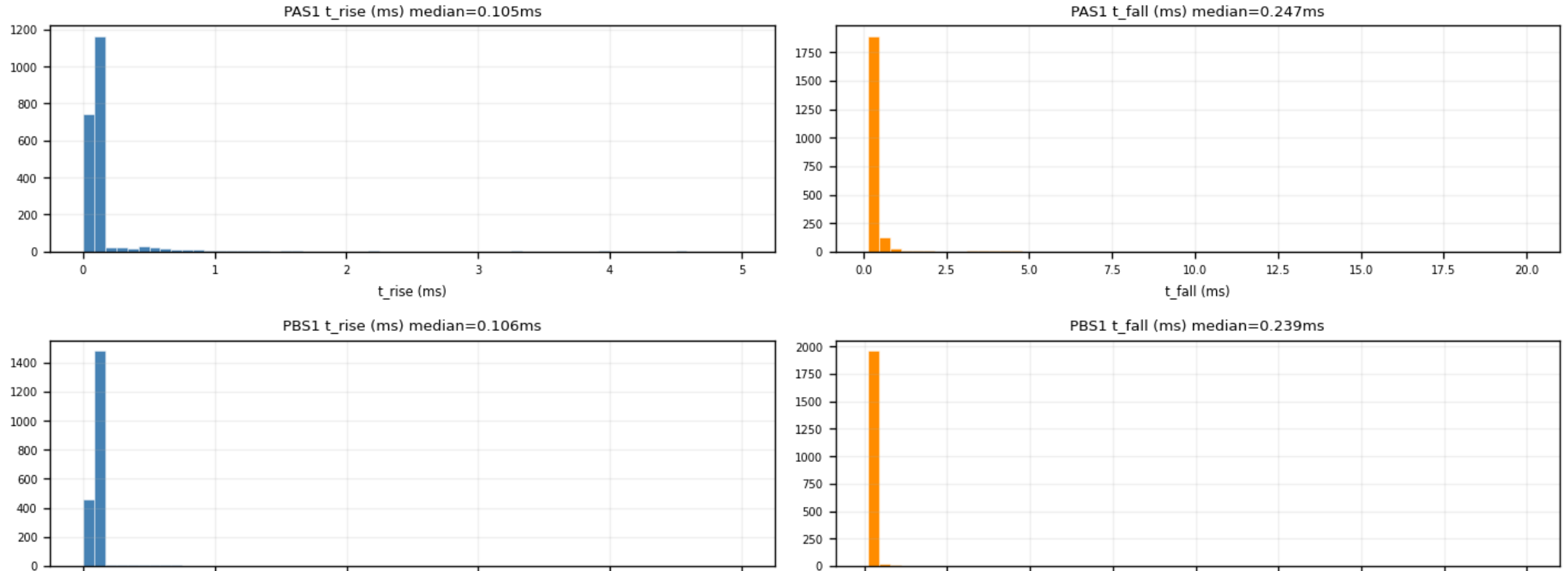
part 2/2 (top to bottom)

Backup — Z7 · time_constants (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: fitted  $t_{\text{rise}}$  /  $t_{\text{fall}}$  distributions (ms) of fit_ok events; free-pretrigger fit, loose bounds  $t_{\text{rise}} < 5\text{ms}$   $t_{\text{fall}} < 20\text{ms}$ 
```

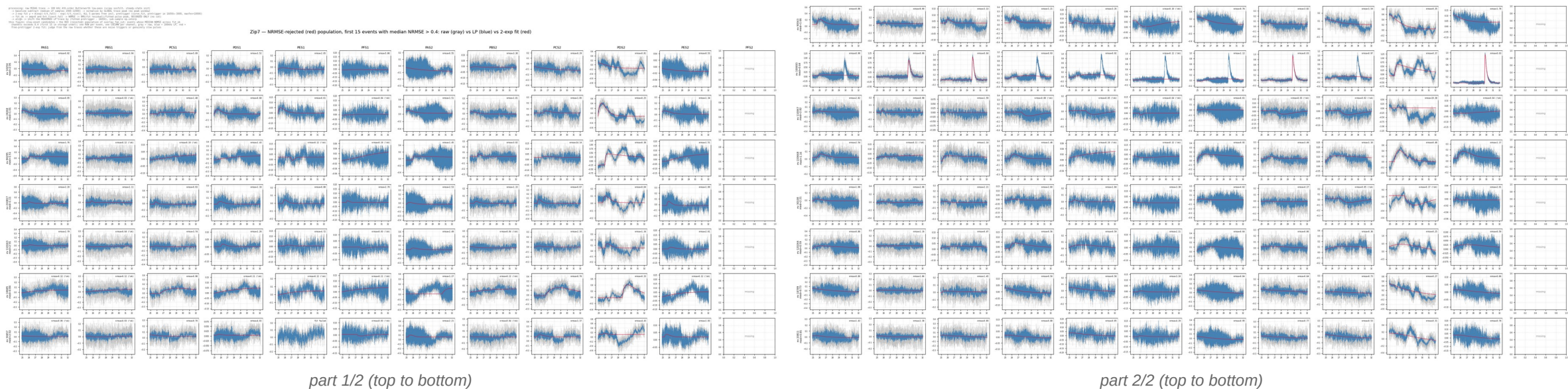
Zip7 — t_{rise} / t_{fall} (fit_ok events, free-pretrigger fit)



Backup — Z7 · slow_rise_events

the NRMSE-rejected population · figure: zip7_slow_rise_events.png

Events whose median NRMSE across channels > 0.4. Raw traces show no pulse in any channel: the rejected population is noise triggers, not physics.

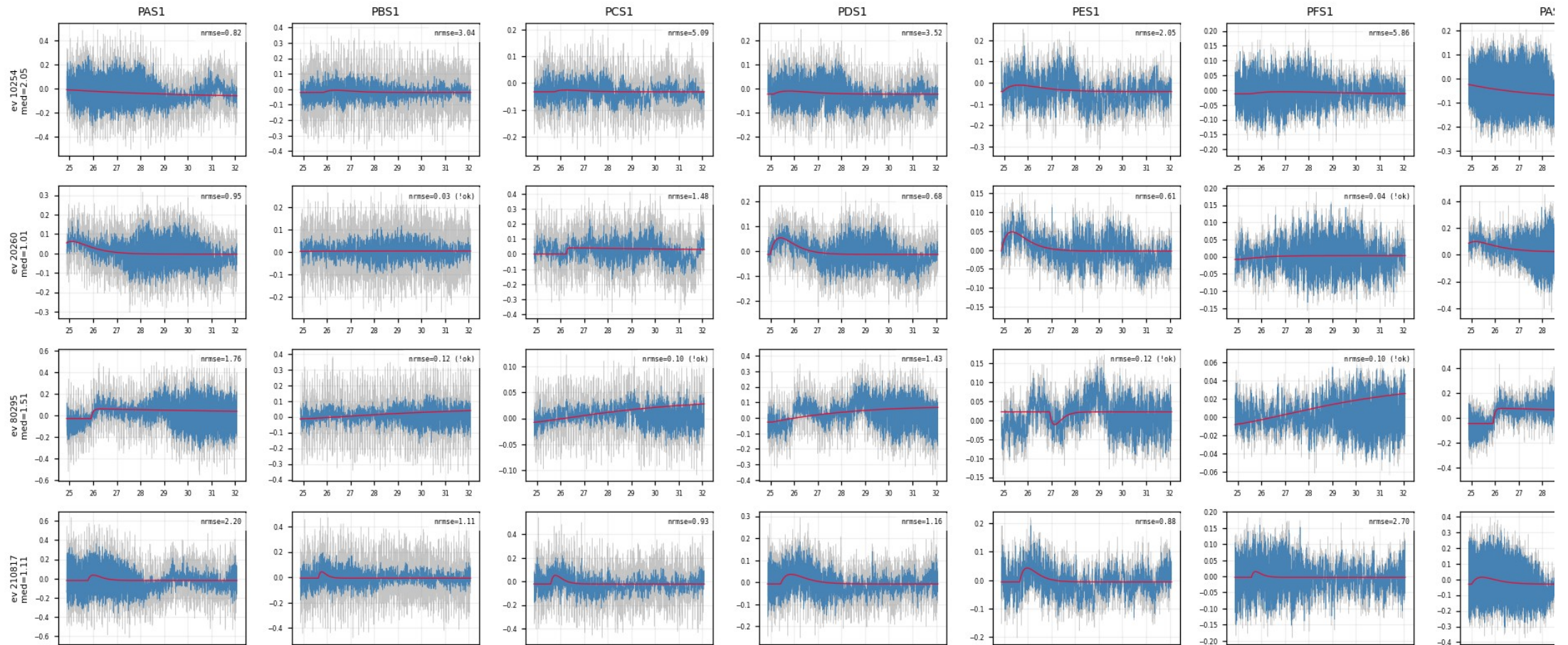


Backup — Z7 · slow_rise_events (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t_rise<fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: slow-onset candidates = the RED (rejected) population of overlay fan cut: events whose MEDIAN NRMSE across fit ok
channels exceeds 0.4 (first 15 in storage order); one ROW per event, one COLUMN per channel, gray = raw, blue = 100kHz LP, red =
free-pretrigger 2-exp fit; judge from the raw traces whether these are noise triggers or genuinely slow pulses
```

Zip7 — NRMSE-rejected (red) population, first 15 events with median NRMSE >

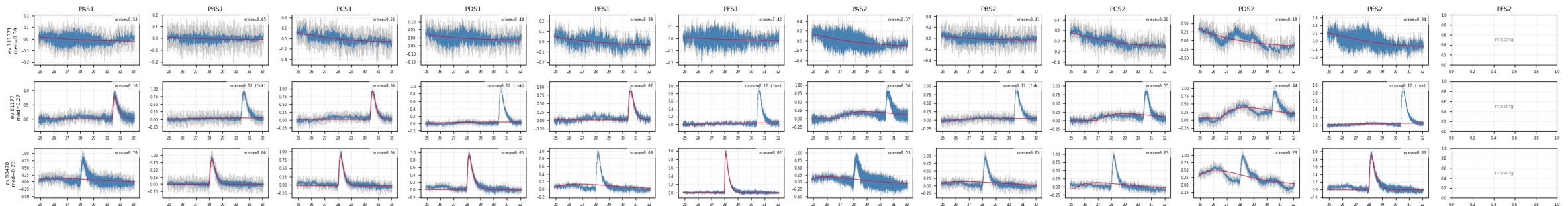


well-fit slow-rise (echo-trigger) events · figure: zip7_shadow_events.png

Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} > 0.2$ ms: genuinely slow, well-fit pulses — the faint displaced bundle seen in aligned overlays; kept, and exactly what NxM multi-templates are for.

```
#processing raw MIDAS trace > 10e4 4th-order Butterworth low-pass [scipy sosfilt, steady-state init]
> baseline subtract [median of samples(10000-12000)] > normalize by GLOBAL trace peak (no peak window)
> 2-exp fit > Atemp, fit = fit(x, y, xerr, yerr, model=fitfunc, p0=[10000, 10000], maxfev=10000)
> fitok = amp && dO_fitset[all] == NMRSE == RMS(fit residual)/fitted pulse peak, RECORDED_ONLY [no cut]
> align > shift the MEASURED LP trace by fit parameter - t650ns, min_sample.no_interfer
> plot aligned position of aligned signal vs time, all-fit events overlaid across fit ok channels < 0.4 Å and
median trise > 0.20 ns (first 3 in storage_order); event aliases at t650ns by the peaks comes later, producing the
displaced bubble in aligned_overlay; one ROW per onset, one COLUMN per channel, gray = raw, blue = 100nA/LP, red = 2-exp fit
```

Zip7 — ghost population (well-fit slow-rise events), first 3 events with median NRMSE ≤ 0.4 and median $t_{\text{rise}} > 0.20$ ms: raw (gray) vs LP (blue) vs 2-exp fit (red)

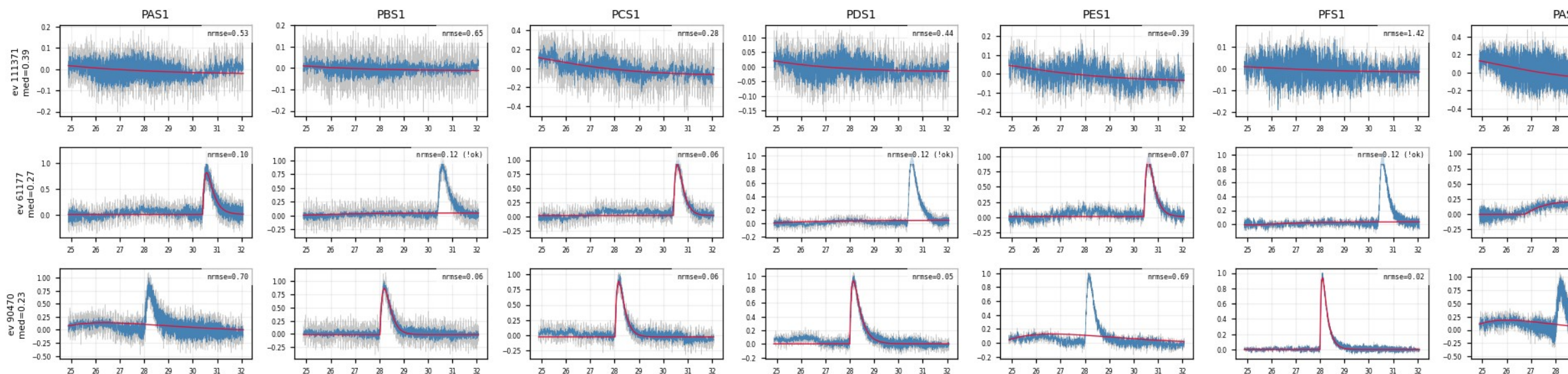


Backup — Z7 · shadow_events (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

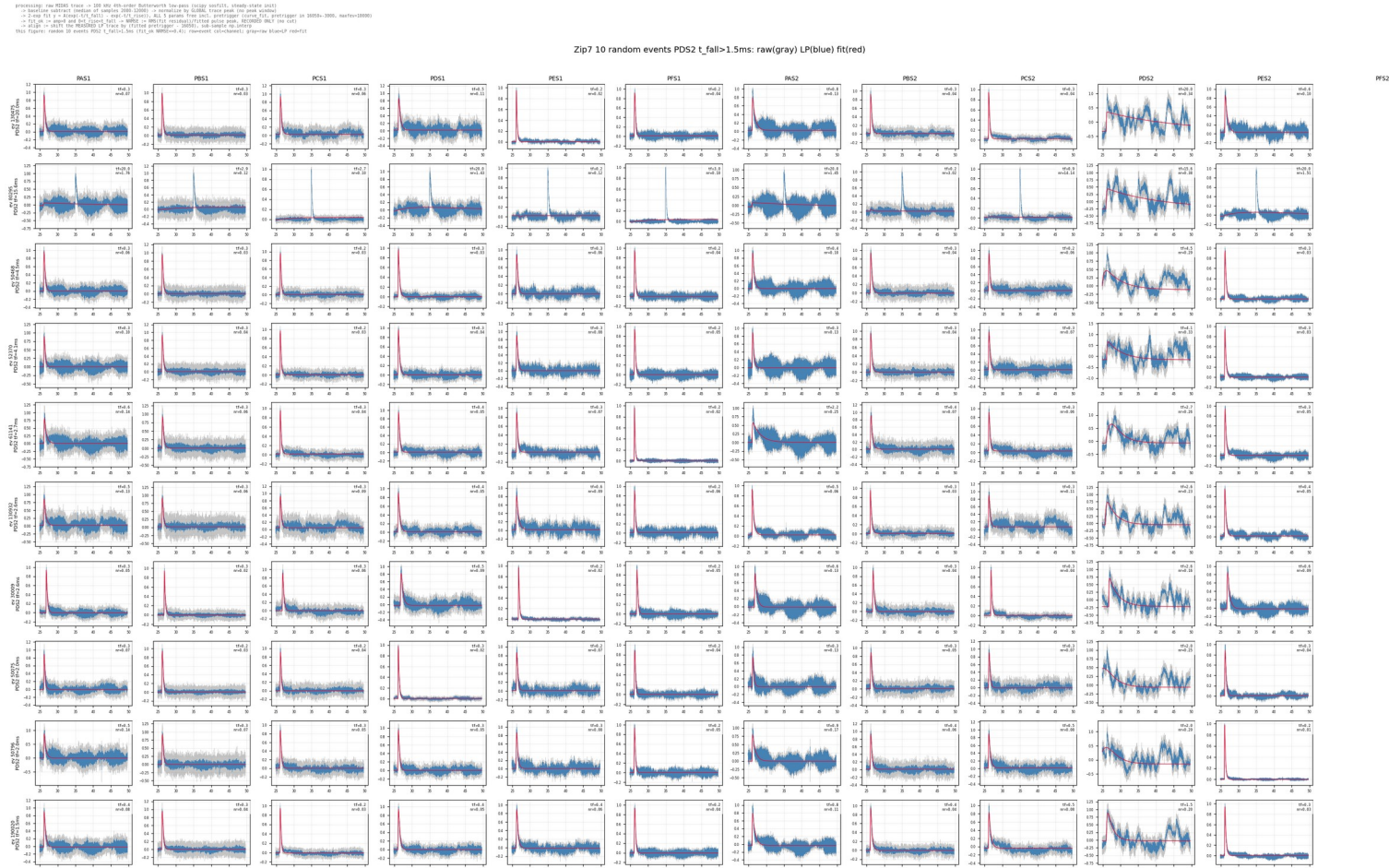
```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t_rise< fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: GHOST population of aligned overlay = genuinely slow, WELL-FIT events: median NRMSE across fit ok channels <= 0.4 AND
median t_rise > 0.20 ms (first 3 in storage order); onset aligns at 16050 but the peak comes later, producing the faint
displaced bundle in aligned_overlay; one ROW per event, one COLUMN per channel, gray = raw, blue = 100kHz LP, red = 2-exp fit
```

Zip7 — ghost population (well-fit slow-rise events), first 3 events with median NRMSE <= 0.4 and n



long- τ fall study (PDS2 reference) · figure: zip7_PDS2_slow_fall.png

10 random events with PDS2 $\tau_{\text{fall}} > 1.5$ ms drawn in every channel: 11 channels show normal fast pulses; the long fall is a PDS2-only low-frequency artifact — not slow physics, no event-level cut.

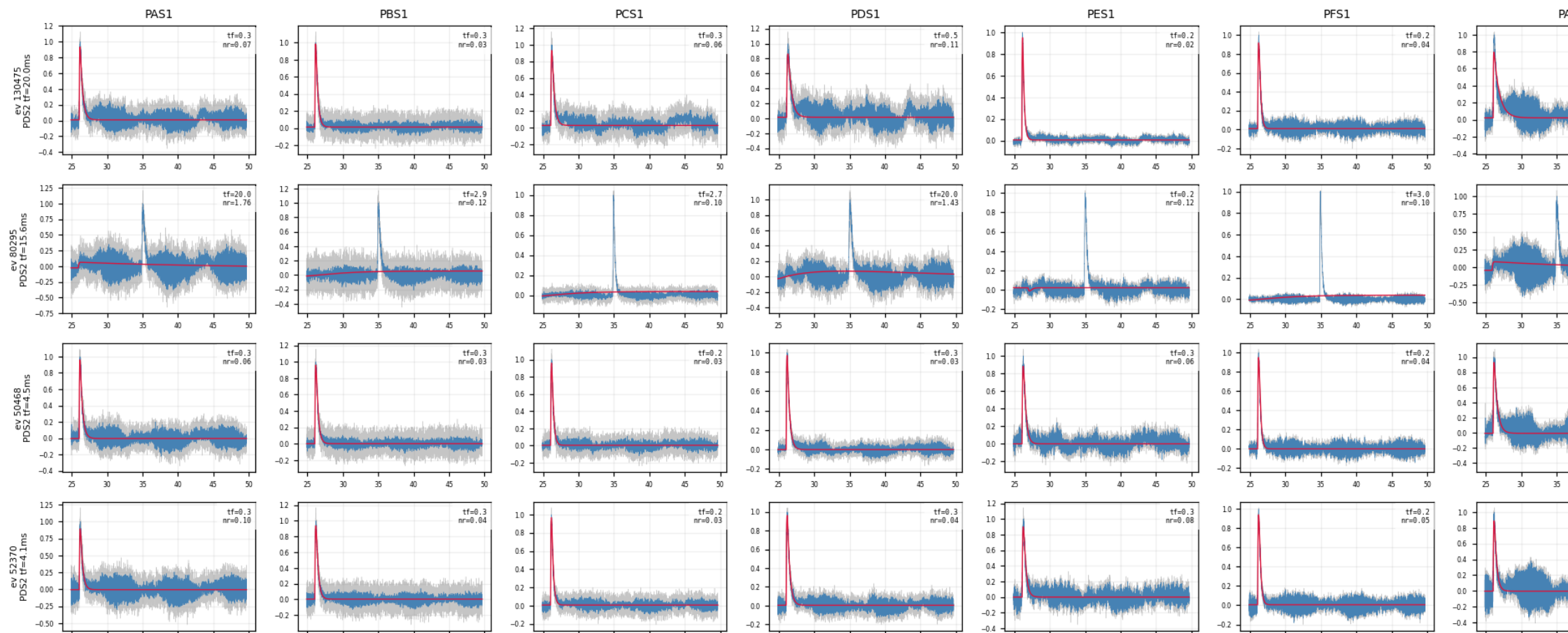


Backup — Z7 · slow_fall_events (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: random 10 events PDS2 t_fall>1.5ms (fit_ok NRMSE<=0.4); row=event col=channel; gray=raw blue=LP red=fit
```

Zip7 10 random events PDS2 t_fall>1.5ms: raw(gray)



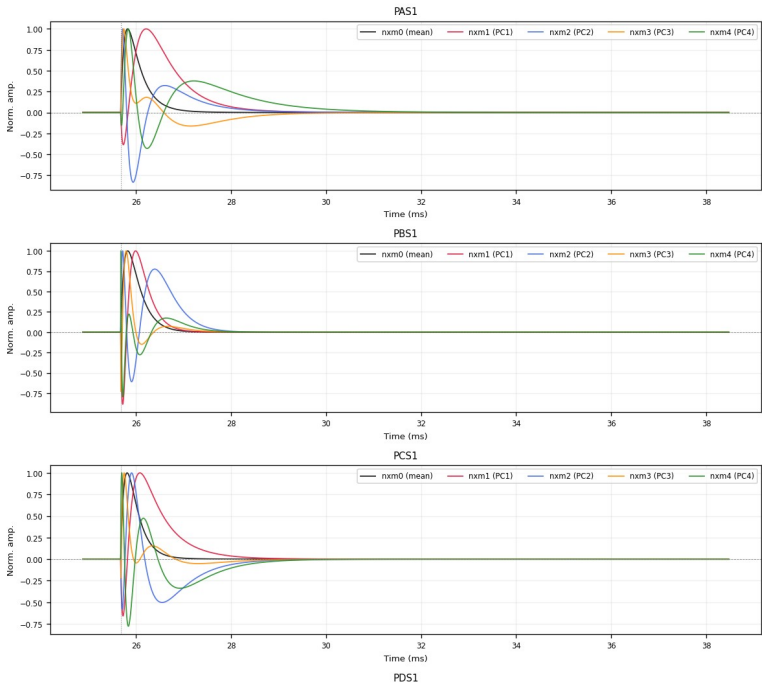
Backup — Z7 · deliverables/nxm

PCA templates nxm0–4 · figure: zip7_pca_templates.png

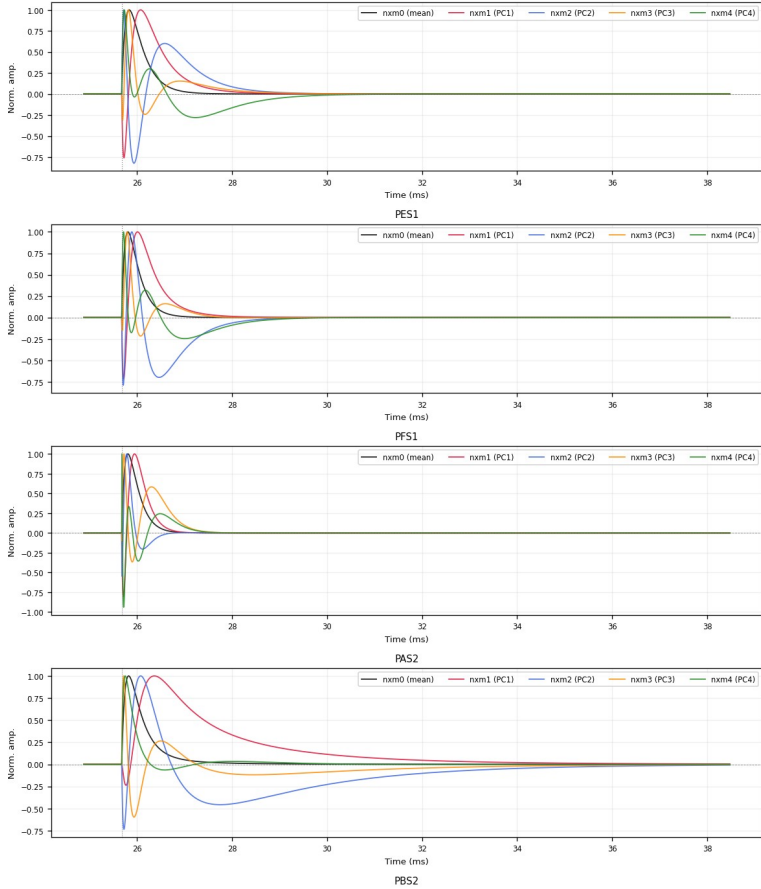
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized to 1; PC1+PC2 capture 96–98% of the shape variance.

processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> Baseline subtract (median of samples 2000-12000) -> normalize by (LSDML trace peak (no peak window))
-> 2-seg fit $y = A(\exp(-t/\tau_{fall}) - \exp(-t/\tau_{rise}))$; ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxrev=10000)
-> Fit ok -> expm and fit ok -> MUSE -> MUSE fit residual (fitted pulse peak, 16050-3000 ONLY (no cut))
-> align -> shift the HEADRED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: Non PCA templates: All peak-normalized to unit peak abs (invd mean + real 4 PCA components 1-4). Population = fitted
2-exp curves with fit ok AND NRMS<0.4 AND $\tau_{rise}<0.30$ ms at common pretrigger 16050; per-channel PCA. Real pulse = sum_i amp_i
+ nxm_i.

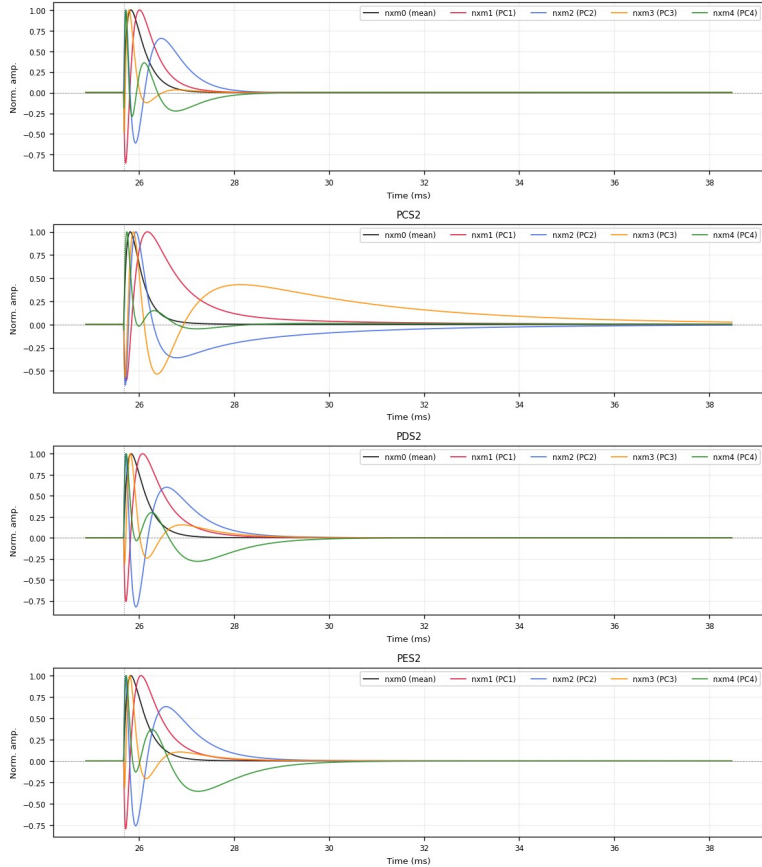
Zip7 - NxM PCA templates (all peak-normalized to 1)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



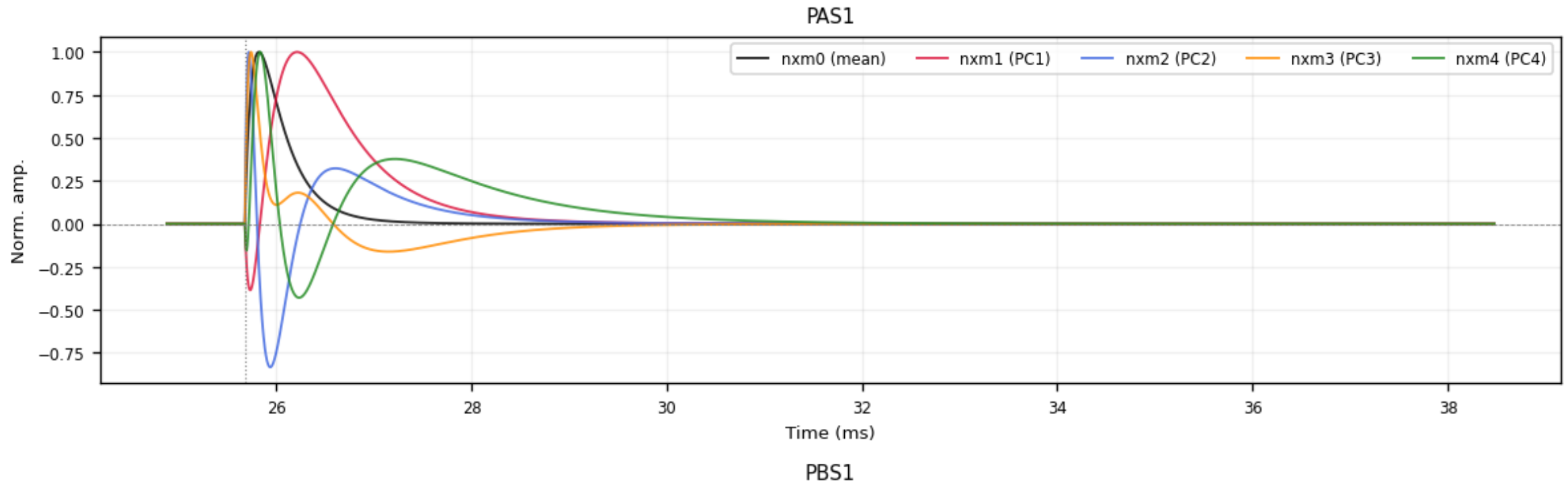
part 3/3 (top to bottom)

Backup — Z7 · deliverables/nxm (detail 1:1)

top region of the same figure at full resolution — whole figure on the previous slide

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NxM PCA templates, ALL peak-normalized to unit peak-abs (nxm0 mean + nxm1-4 PCA components 1-4). Population = fitted
2-exp curves with fit_ok AND NRMSE<=0.4 AND t_rise<=0.30ms at common pretrigger 16050; per-channel PCA. Real pulse = sum_i amp_i
* nxm_i.
```

Zip7 - NxM PCA templates (all peak-normalized to 1)

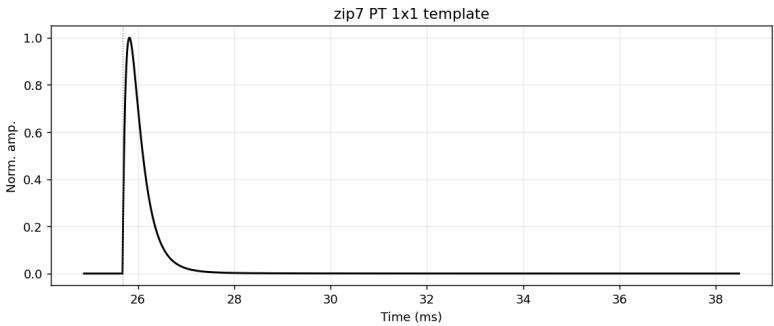


Backup — Z7 · 1x1 summed templates PT / PS1 / PS2

figures: deliverables/1x1/plots/{PT,PS1,PS2}/zip7_*.png

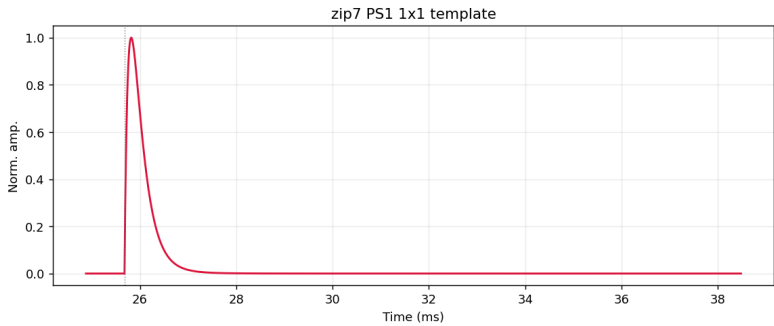
Single-template (1x1) summed curves: peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and B<t_rise<fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PT 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (all channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



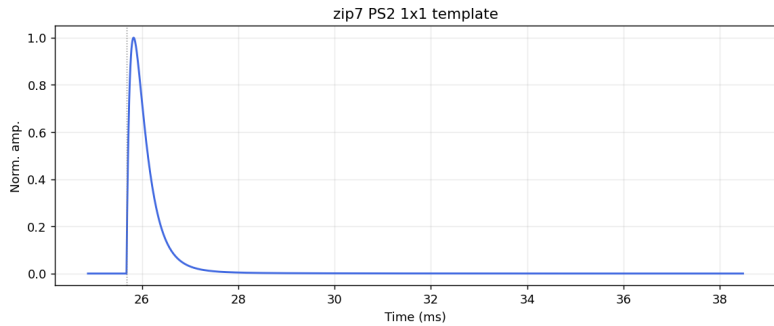
Z7 PT

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and B<t_rise<fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PS1 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (side-1 channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



Z7 PS1

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and B<t_rise<fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PS2 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (side-2 channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



Z7 PS2